CSE370 : Database Systems
Project Report
Project Title : Credit Risk and Loan Monitoring System

Group No : __, CSE370 Lab Section : __, Spring 2026		
ID	**Name**	**Contribution**
23301666	Tasnuba Tabassum Ahona	ALL

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Introduction

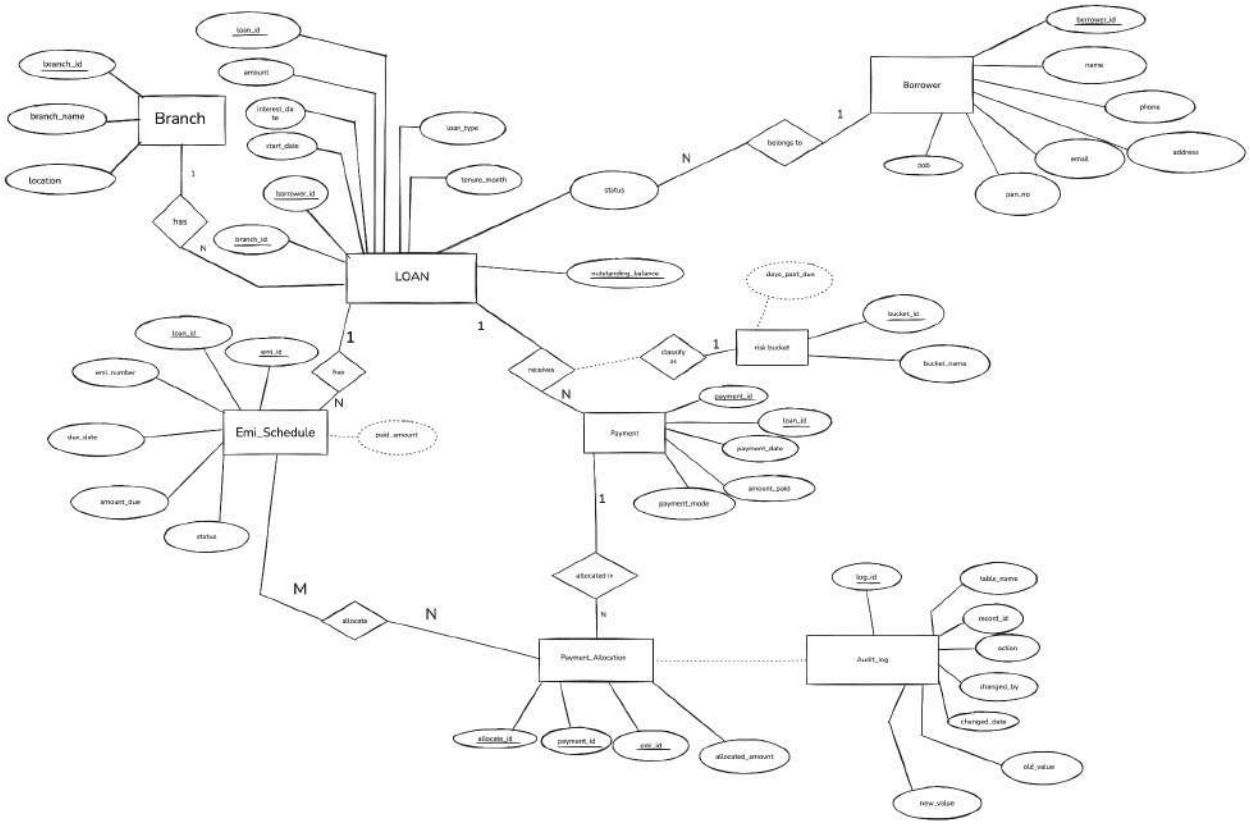
This project is a Credit Risk and Loan Monitoring Platform designed as a simplified banking system. It facilitates interaction between three main roles: Borrowers who request loans, Staff who approve or reject requests, and Admins who oversee system operations. The core functionality revolves around tracking loan repayments through monthly Equated Monthly Installment (EMI) schedules and automatically classifying loans into risk buckets (e.g., Current, 30+ Days, 90+ Days) based on payment delays.

Project Features

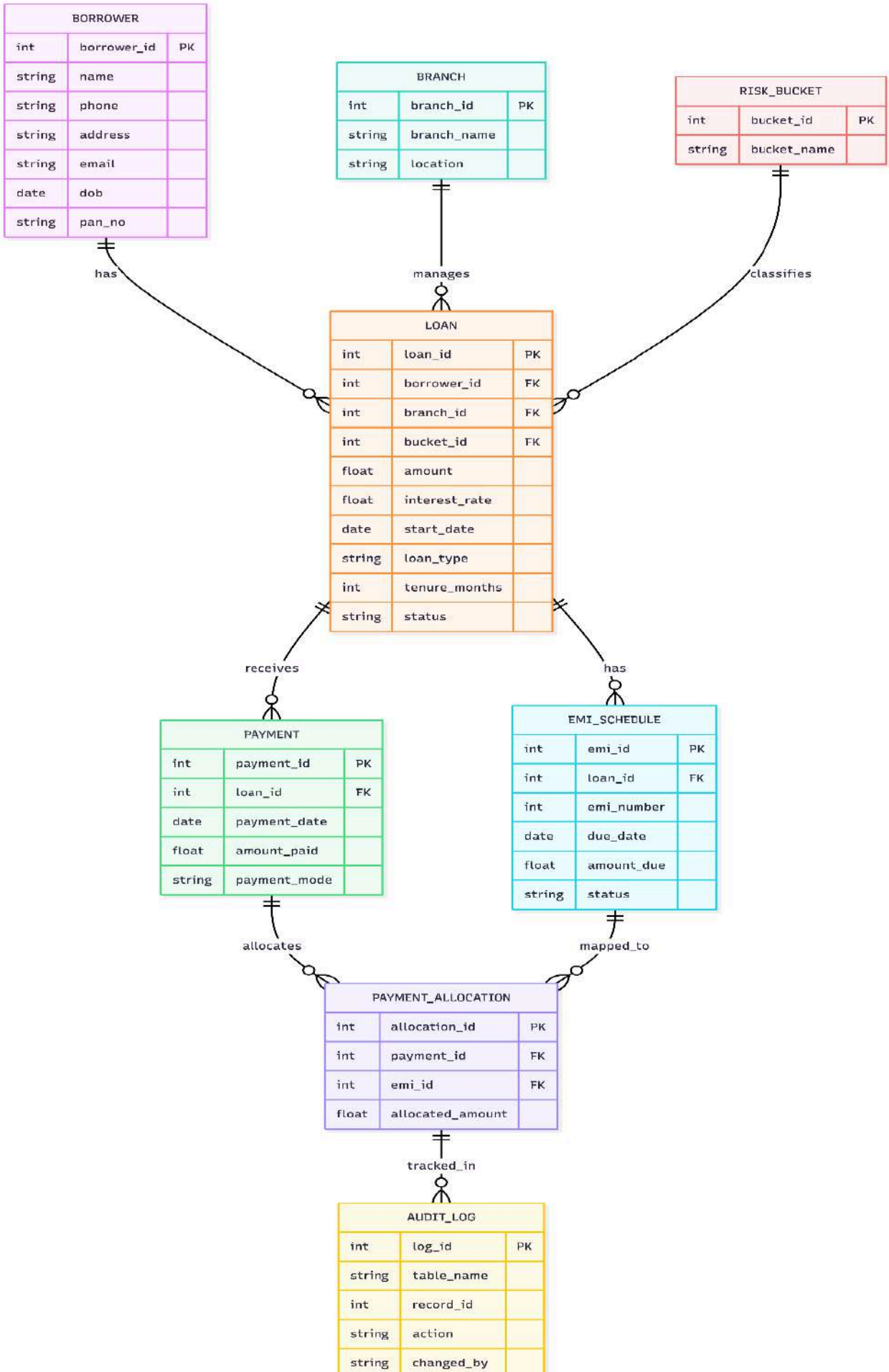
ID, Name	Features [3 per member]	
23301666, Tasnuba Tabassum Ahona	Ft 1	**Role-Based Access Control:** Separate interfaces and permissions for Admins, Staff, and Borrowers using secure session management.
	Ft 2	**Automated EMI Generation:** Upon loan approval, the system automatically calculates simple interest and generates a

		complete repayment schedule.
	Ft 3	**Dynamic Risk Classification:** Real-time calculation of loan risk based on the number of days the oldest unpaid EMI is overdue.
	Ft 4	**Flexible Payment Allocation:** A single payment from a borrower can be allocated across multiple pending EMIs.
	Ft 5	**Comprehensive Audit Logging:** Every significant system action (INSERT, UPDATE, DELETE, APPROVE) is recorded for accountability.
	Ft 6	**Branch-Scoped Data Isolation:** Staff members are restricted to viewing and managing only the loans associated with their specific branch.

ER/EER Diagram



Schema Diagram


BORROWER
int borrower_id PK
string name
string phone
string address
string email
date dob
string pan_no
BRANCH
int branch_id PK
string branch_name
string location
RISK_BUCKET
int bucket_id PK
string bucket_name
has
manages
classifies
LOAN
int loan_id PK
int borrower_id FK
int branch_id FK
int bucket_id FK
float amount
float interest_rate
date start_date
string loan_type
int tenure_months
string status
receives
has
PAYMENT
int payment_id PK
int loan_id FK
date payment_date
float amount_paid
string payment_mode
EMI_SCHEDULE
int emi_id PK
int loan_id FK
int emi_number
date due_date
float amount_due
string status
allocates
mapped_to
PAYMENT_ALLOCATION
int allocation_id PK
int payment_id FK
int emi_id FK
float allocated_amount
tracked_in
AUDIT_LOG
int log_id PK
string table_name
int record_id
string action
string changed_by

Normalization

a. First Normal Form (1NF)

The schema is in 1NF.

- Each table (e.g., User, Loan, Borrower) contains a unique Primary Key.
- All columns contain atomic values; there are no multi-valued attributes or repeating groups (for example, the Borrower table stores separate columns for phone, email, and address).

b. Second Normal Form (2NF)

The schema is in **2NF**.

- A table is in 2NF if it is in 1NF and contains no **partial functional dependencies**, meaning non-prime attributes must depend on the *entire* primary key.
- In this schema, most tables use a single-column primary key (like `user_id` or `loan_id`), which mathematically eliminates the possibility of partial dependencies. Even in mapping tables like `Payment_Allocation`, attributes depend on the unique `allocation_id`.

c. Third Normal Form (3NF)

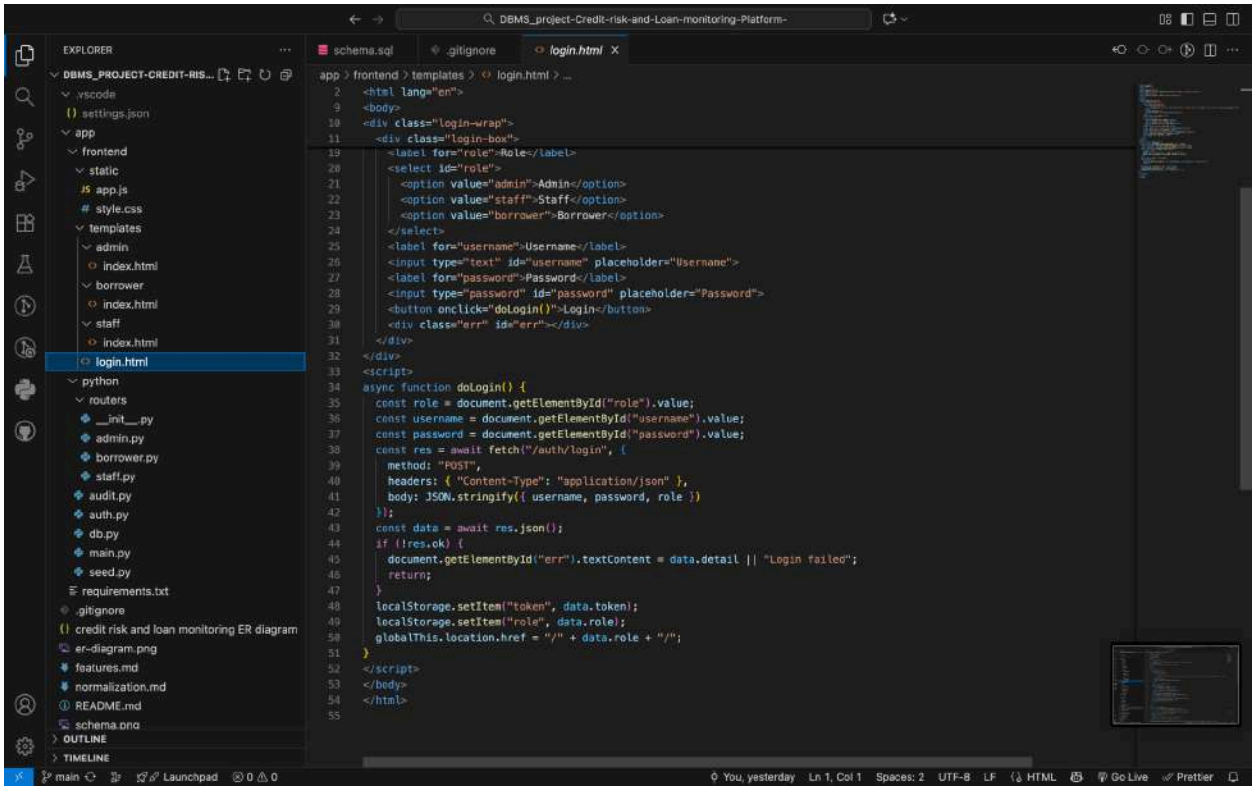
The schema is in **3NF**.

- A schema is in 3NF if it has no **transitive dependencies** (where a non-key attribute depends on another non-key attribute).
- This design ensures direct dependency by using **Foreign Keys**. For example, the `Borrower` table stores a `branch_id` instead of branch details like name or location. To find the branch name, a JOIN is performed with the `Branch` table, ensuring `branch_name` depends only on `branch_id` and not transitively on `borrower_id`.

Frontend Development

The frontend was developed using HTML templates rendered by Jinja2.

- Individual Contribution (23301666): I designed and implemented the dashboard for Borrowers to view their active loans and repayment schedules, and the Admin interface for managing the bank's branches and staff accounts.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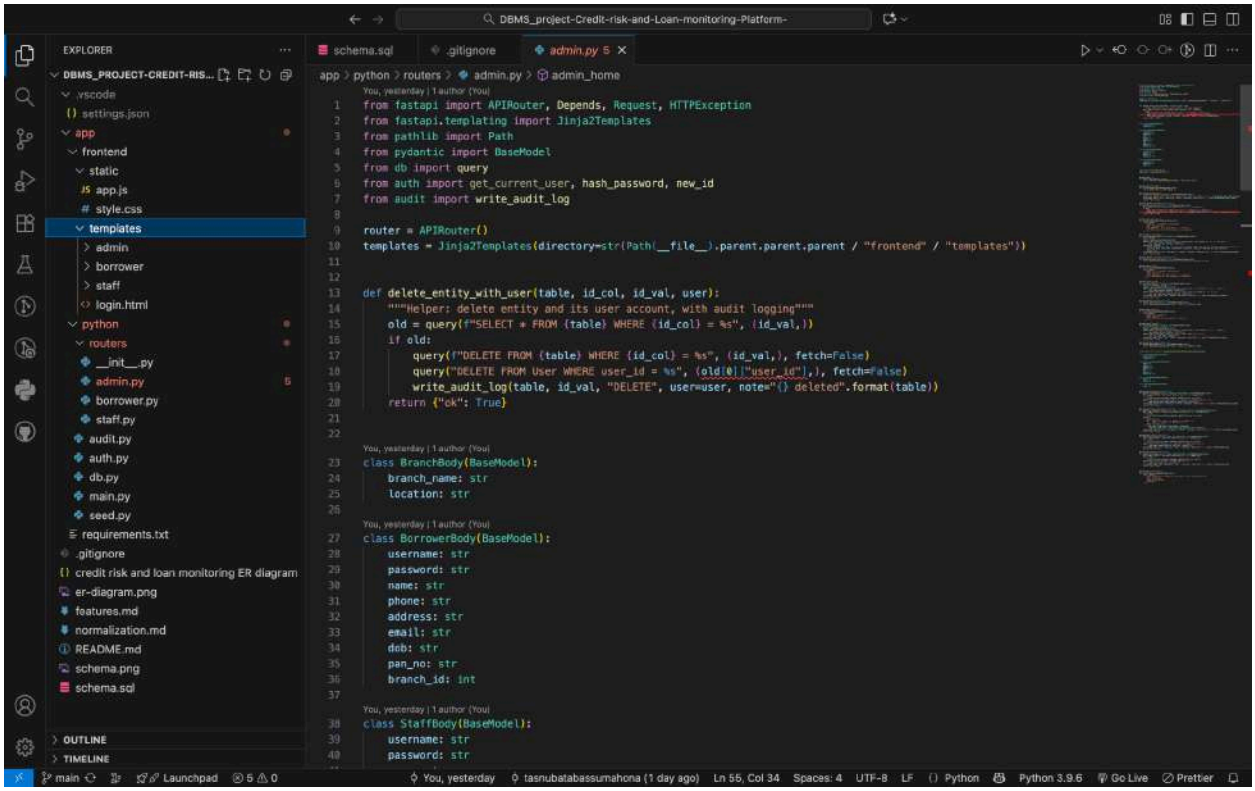
2 <html lang="en">
9 <body>
10
11
19 <label for="role">Role</label>
20 <select id="role">
21 <option value="admin">Admin</option>
22 <option value="staff">Staff</option>
23 <option value="borrower">Borrower</option>
24 </select>
25 <label for="username">Username</label>
26 <input type="text" id="username" placeholder="Username">
27 <label for="password">Password</label>
28 <input type="password" id="password" placeholder="Password">
29 <button onclick="doLogin()">Login</button>
30
31
32
33 <script>
34 async function doLogin() {
35 const role = document.getElementById("role").value;
36 const username = document.getElementById("username").value;
37 const password = document.getElementById("password").value;
38 const res = await fetch("/auth/login", {
39 method: "POST",
40 headers: { "Content-Type": "application/json" },
41 body: JSON.stringify({ username, password, role })
42 });
43 const data = await res.json();
44 if (!res.ok) {
45 document.getElementById("err").textContent = data.detail || "Login failed";
46 return;
47 }
48 localStorage.setItem("token", data.token);
49 localStorage.setItem("role", data.role);
50 globalThis.location.href = "/" + data.role + "/";
51 }
52 </script>
53 </body>
54 </html>
55
main
Launchpad
0 0
You, yesterday
Ln 1, Col 1
Spaces: 2
UTF-8
LF
HTML
Go Live
Prettier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app > python > routers > admin.py > admin_home
You, yesterday | 1 author (You)
1 from fastapi import APIRouter, Depends, Request, HTTPException
2 from fastapi.templating import Jinja2Templates
3 from pathlib import Path
4 from pydantic import BaseModel
5 from db import query
6 from auth import get_current_user, hash_password, new_id
7 from audit import write_audit_log
8
9 router = APIRouter()
10 templates = Jinja2Templates(directory=str(Path(__file__).parent.parent.parent / "frontend" / "templates"))
11
12
13 def delete_entity_with_user(table, id_col, id_val, user):
14 """Helper: delete entity and its user account, with audit logging"""
15 old = query(f"SELECT * FROM {table} WHERE {id_col} = %s", (id_val,))
16 if old:
17 query(f"DELETE FROM {table} WHERE {id_col} = %s", (id_val,), fetch=False)
18 query("DELETE FROM User WHERE user_id = %s", (old[0]["user_id"],), fetch=False)
19 write_audit_log(table, id_val, "DELETE", user=user, note="{} deleted".format(table))
20 return {"ok": True}
21
22
You, yesterday | 1 author (You)
23 class BranchBody(BaseModel):
24 branch_name: str
25 location: str
26
You, yesterday | 1 author (You)
27 class BorrowerBody(BaseModel):
28 username: str
29 password: str
30 name: str
31 phone: str
32 address: str
33 email: str
34 dob: str
35 pan_no: str
36 branch_id: int
37
You, yesterday | 1 author (You)
38 class StaffBody(BaseModel):
39 username: str
40 password: str
main
Launchpad
5 0
You, yesterday
tasnubatabassumahona (1 day ago)
Ln 55, Col 34
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Prettier

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app > python > routers > admin.py > admin_home

```python
You, yesterday | 1 author (You)
class AdminBody(BaseModel):
    username: str
    password: str
    name: str


from auth import require_role
admin_only = require_role("admin")


@router.get("/")
def admin_home(request: Request):
    return templates.TemplateResponse(request, "admin/index.html")


@router.get("/branches")
def list_branches(user=Depends(admin_only)):
    return query("SELECT * FROM Branch")

@router.post("/branches")
def create_branch(body: BranchBody, user=Depends(admin_only)):
    branch_id = query("INSERT INTO Branch (branch_name, location) VALUES (%s, %s)", (body.branch_name, body.location), fetc
    write_audit_log("Branch", branch_id, "INSERT", user=user, note="Branch '{}' added".format(body.branch_name))
    return {"ok": True}

@router.delete("/branches/{branch_id}")
def delete_branch(branch_id: int, user=Depends(admin_only)):
    old = query("SELECT * FROM Branch WHERE branch_id = %s", (branch_id,))
    if old:
        query("DELETE FROM Branch WHERE branch_id = %s", (branch_id,), fetch=False)
        write_audit_log("Branch", branch_id, "DELETE", user=user, note="Branch '{}' deleted".format(old[0].get("branch_name
    return {"ok": True}


@router.get("/borrowers")
def list_borrowers(user=Depends(admin_only)):
    return query("""
        SELECT b.*, u.username, br.branch_name
        FROM   Borrower AS b
        LEFT JOIN User   AS u  ON b.user_id   = u.user_id
        LEFT JOIN Branch AS br ON b.branch_id = br.branch_id
```

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app > python > routers > admin.py > admin_home

```python
@router.post("/borrowers")
def create_borrower(body: BorrowerBody, user=Depends(admin_only)):
    uid = new_id()
    hashed = hash_password(body.password)
    query("INSERT INTO User (user_id, username, password_hash, role) VALUES (%s, %s, %s, 'borrower')",
          (uid, body.username, hashed), fetch=False)
    borrower_id = query(
        "INSERT INTO Borrower (user_id, name, phone, address, email, dob, pan_no, branch_id) VALUES (%s,%s,%s,%s,%s,%s,%s,%
        (uid, body.name, body.phone, body.address, body.email, body.dob, body.pan_no, body.branch_id),
        fetch=False)
    write_audit_log("Borrower", borrower_id, "INSERT", user=user, note="Borrower '{}' added".format(body.username))
    return {"ok": True}

@router.delete("/borrowers/{borrower_id}")
def delete_borrower(borrower_id: int, user=Depends(admin_only)):
    return delete_entity_with_user("Borrower", "borrower_id", borrower_id, user)


@router.get("/staff")
def list_staff(user=Depends(admin_only)):
    return query("""
        SELECT s.*, u.username, b.branch_name
        FROM   Staff AS s
        INNER JOIN User   AS u ON s.user_id   = u.user_id
        LEFT  JOIN Branch AS b ON s.branch_id = b.branch_id
    """)

@router.post("/staff")
def create_staff(body: StaffBody, user=Depends(admin_only)):
    uid, sid = new_id(), new_id()
    hashed = hash_password(body.password)
    query("INSERT INTO User (user_id, username, password_hash, role) VALUES (%s, %s, %s, 'staff')",
          (uid, body.username, hashed), fetch=False)
    query("INSERT INTO Staff (staff_id, user_id, name, branch_id) VALUES (%s, %s, %s, %s)",
          (sid, uid, body.name, body.branch_id), fetch=False)
    write_audit_log("Staff", sid, "INSERT", user=user, note="Staff '{}' added".format(body.username))
    return {"ok": True}

@router.delete("/staff/{staff_id}")
def delete_staff(staff_id: str, user=Depends(admin_only)):
    return delete_entity_with_user("Staff", "staff_id", staff_id, user)
```

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```python
@router.get("/admins")
def list_admins(user=Depends(admin_only)):
    return query("""
        SELECT a.*, u.username
        FROM   Admin AS a
        INNER JOIN User AS u ON a.user_id = u.user_id
    """)

@router.post("/admins")
def create_admin(body: AdminBody, user=Depends(admin_only)):
    uid, aid = new_id(), new_id()
    hashed = hash_password(body.password)
    query("INSERT INTO User (user_id, username, password_hash, role) VALUES (%s, %s, %s, 'admin')",
          (uid, body.username, hashed), fetch=False)
    query("INSERT INTO Admin (admin_id, user_id, name) VALUES (%s, %s, %s)",
          (aid, uid, body.name), fetch=False)
    write_audit_log("Admin", aid, "INSERT", user=user, note="Admin '{}' added".format(body.username))
    return {"ok": True}

@router.delete("/admins/{admin_id}")
def delete_admin(admin_id: str, user=Depends(admin_only)):
    return delete_entity_with_user("Admin", "admin_id", admin_id, user)


# — Update (PUT) endpoints ————————————————————————————————

You, yesterday | 1 author (You)
class BranchUpdateBody(BaseModel):
    branch_name: str
    location: str

You, yesterday | 1 author (You)
class BorrowerUpdateBody(BaseModel):
    name: str
    phone: str
    address: str
    email: str
    dob: str
    pan_no: str
    branch_id: int

You, yesterday | 1 author (You)
class StaffUpdateBody(BaseModel):
```

main | Launchpad | ⊗ 5 ⚠ 0 | You, yesterday | tasnubatabassumahona (1 day ago) | Ln 55, Col 34 | Spaces: 4 | UTF-8 | LF | {} Python | Python 3.9.6 | Go Live | Prettier

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```python
@router.get("/admins")
def list_admins(user=Depends(admin_only)):
    return query("""
        SELECT a.*, u.username
        FROM   Admin AS a
        INNER JOIN User AS u ON a.user_id = u.user_id
    """)

@router.post("/admins")
def create_admin(body: AdminBody, user=Depends(admin_only)):
    uid, aid = new_id(), new_id()
    hashed = hash_password(body.password)
    query("INSERT INTO User (user_id, username, password_hash, role) VALUES (%s, %s, %s, 'admin')",
          (uid, body.username, hashed), fetch=False)
    query("INSERT INTO Admin (admin_id, user_id, name) VALUES (%s, %s, %s)",
          (aid, uid, body.name), fetch=False)
    write_audit_log("Admin", aid, "INSERT", user=user, note="Admin '{}' added".format(body.username))
    return {"ok": True}

@router.delete("/admins/{admin_id}")
def delete_admin(admin_id: str, user=Depends(admin_only)):
    return delete_entity_with_user("Admin", "admin_id", admin_id, user)


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    name: str
    phone: str
    address: str
    email: str
    dob: str
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```

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```python
@router.put("/branches/{branch_id}")
def update_branch(branch_id: int, body: BranchUpdateBody, user=Depends(admin_only)):
    old = query("SELECT * FROM Branch WHERE branch_id = %s", (branch_id,))
    if not old:
        raise HTTPException(status_code=404, detail="Branch not found")
    query("UPDATE Branch SET branch_name = %s, location = %s WHERE branch_id = %s",
          (body.branch_name, body.location, branch_id), fetch=False)
    write_audit_log("Branch", branch_id, "UPDATE", user=user, note="Branch '{}' updated".format(body.branch_name))
    return {"ok": True}

@router.put("/borrowers/{borrower_id}")
def update_borrower(borrower_id: int, body: BorrowerUpdateBody, user=Depends(admin_only)):
    old = query("SELECT * FROM Borrower WHERE borrower_id = %s", (borrower_id,))
    if not old:
        raise HTTPException(status_code=404, detail="Borrower not found")
    query("""
        UPDATE Borrower
        SET    name = %s, phone = %s, address = %s, email = %s,
               dob = %s, pan_no = %s, branch_id = %s
        WHERE  borrower_id = %s
    """, (body.name, body.phone, body.address, body.email,
          body.dob, body.pan_no, body.branch_id, borrower_id), fetch=False)
    write_audit_log("Borrower", borrower_id, "UPDATE", user=user, note="Borrower '{}' updated".format(body.name))
    return {"ok": True}

@router.put("/staff/{staff_id}")
def update_staff(staff_id: str, body: StaffUpdateBody, user=Depends(admin_only)):
    old = query("SELECT * FROM Staff WHERE staff_id = %s", (staff_id,))
    if not old:
        raise HTTPException(status_code=404, detail="Staff not found")
    query("UPDATE Staff SET name = %s, branch_id = %s WHERE staff_id = %s",
          (body.name, body.branch_id, staff_id), fetch=False)
    write_audit_log("Staff", staff_id, "UPDATE", user=user, note="Staff '{}' updated".format(body.name))
    return {"ok": True}

@router.put("/admins/{admin_id}")
def update_admin(admin_id: str, body: AdminUpdateBody, user=Depends(admin_only)):
    old = query("SELECT * FROM Admin WHERE admin_id = %s", (admin_id,))
    if not old:
        raise HTTPException(status_code=404, detail="Admin not found")
    query("UPDATE Admin SET name = %s WHERE admin_id = %s",
          (body.name, admin_id), fetch=False)
    write_audit_log("Admin", admin_id, "UPDATE", user=user, note="Admin '{}' updated".format(body.name))
    return {"ok": True}
```

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    if not old:
        raise HTTPException(status_code=404, detail="Admin not found")
    query("UPDATE Admin SET name = %s WHERE admin_id = %s",
          (body.name, admin_id), fetch=False)
    write_audit_log("Admin", admin_id, "UPDATE", user=user, note="Admin '{}' updated".format(body.name))
    return {"ok": True}


@router.get("/audit-logs")
def audit_logs(user=Depends(admin_only)):
    return query("""
        SELECT change_date, changed_by, table_name, action, new_value AS note
        FROM   Audit_Log
        ORDER BY log_id DESC
        LIMIT 200
    """)
```

main | Launchpad | ⊗ 5 ⚠ 0 | You, yesterday | tasnubatabassumahona (1 day ago) | Ln 55, Col 34 | Spaces: 4 | UTF-8 | LF | Python | Python 3.9.6 | Go Live | Prettier

DBMS_project-Credit-risk-and-Loan-monitoring-Platform-

EXPLORER

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- .vscode
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OUTLINE

TIMELINE

schema.sql | .gitignore | index.html

app > frontend > templates > admin > index.html > ...

You, yesterday | 1 author (You)

```html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Admin — Loan Manager</title>
<link rel="stylesheet" href="/static/style.css">
</head>
<body>
<div class="layout">
  <div class="sidebar">
    <h3>Admin Panel</h3>
    <a href="#" id="nav-branches" onclick="show('branches')">
      <svg viewBox="0 0 24 24"><path d="M3 9l9-7 9 7v11a2 2 0 01-2 2H5a2 2 0 01-2-2z"/><polyline points="9 22 9 12 15 12 15
    </a>
    <a href="#" id="nav-borrowers" onclick="show('borrowers')">
      <svg viewBox="0 0 24 24"><path d="M20 21v-2a4 4 0 00-4-4H8a4 4 0 00-4 4v2"/><circle cx="12" cy="7" r="4"/></svg>Borro
    </a>
    <a href="#" id="nav-staff" onclick="show('staff')">
      <svg viewBox="0 0 24 24"><path d="M17 21v-2a4 4 0 00-4-4H5a4 4 0 00-4 4v2"/><circle cx="9" cy="7" r="4"/><path d="M23
    </a>
    <a href="#" id="nav-admins" onclick="show('admins')">
      <svg viewBox="0 0 24 24"><path d="M12 22s0-4 8-10V5l-8-3-8 3v7c0 6 8 10 8 10z"/></svg>Admins
    </a>
    <a href="#" id="nav-audit" onclick="show('audit')">
      <svg viewBox="0 0 24 24"><path d="M9 5H7a2 2 0 00-2 2v12a2 2 0 002 2h10a2 2 0 002-2V7a2 2 0 00-2-2h-2"/><rect x="9" y
    </a>
    <button class="logout" onclick="logout()">
      <svg viewBox="0 0 24 24" style="width:14px;height:14px;stroke:currentColor;fill:none;stroke-width:2;vertical-align:mi
    </button>
  </div>
  <div class="content">

    <!— Branches —>
    <div id="sec-branches" class="section" style="display:none">
      <div class="section-header">
        <h2>Branches</h2>
        <button class="btn-add" onclick="openAddBranch()">
          <svg viewBox="0 0 24 24" style="width:13px;height:13px;stroke:currentColor;fill:none;stroke-width:2.5"><line x1="
          New Branch
        </button>
      </div>
      <div id="branches-table"></div>
```

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OUTLINE

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schema.sql | .gitignore | index.html

app > frontend > templates > admin > index.html > ...

```html
<html lang="en">
<body>
<div class="layout">
  <div class="content">

    <!— Borrowers —>
    <div id="sec-borrowers" class="section" style="display:none">
      <div class="section-header">
        <h2>Borrowers</h2>
        <button class="btn-add" onclick="openAddBorrower()">
          <svg viewBox="0 0 24 24" style="width:13px;height:13px;stroke:currentColor;fill:none;stroke-width:2.5"><line x1="
          New Borrower
        </button>
      </div>
      <div id="borrowers-table"></div>
    </div>

    <!— Staff —>
    <div id="sec-staff" class="section" style="display:none">
      <div class="section-header">
        <h2>Staff</h2>
        <button class="btn-add" onclick="openAddStaff()">
          <svg viewBox="0 0 24 24" style="width:13px;height:13px;stroke:currentColor;fill:none;stroke-width:2.5"><line x1="
          New Staff Member
        </button>
      </div>
      <div id="staff-table"></div>
    </div>

    <!— Admins —>
    <div id="sec-admins" class="section" style="display:none">
      <div class="section-header">
        <h2>Admins</h2>
        <button class="btn-add" onclick="openAddAdmin()">
          <svg viewBox="0 0 24 24" style="width:13px;height:13px;stroke:currentColor;fill:none;stroke-width:2.5"><line x1="
          New Admin
        </button>
      </div>
      <div id="admins-table"></div>
    </div>

    <!— Audit Log —>
    <div id="sec-audit" class="section" style="display:none">
```

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schema.sql | .gitignore | index.html

app > frontend > templates > admin > index.html > ...

```html
<html lang="en">
<body>
<div class="layout">
  <div class="content">
    </div>

    <!-- Admins -->
    <div id="sec-admins" class="section" style="display:none">
      <div class="section-header">
        <h2>Admins</h2>
        <button class="btn-add" onclick="openAddAdmin()">
          <svg viewBox="0 0 24 24" style="width:13px;height:13px;stroke:currentColor;fill:none;stroke-width:2.5"><line x1="
          New Admin
        </button>
      </div>
      <div id="admins-table"></div>
    </div>

    <!-- Audit Log -->
    <div id="sec-audit" class="section" style="display:none">
      <h2>Audit Log</h2>
      <p class="section-meta">Most recent 300 events — all create, update and delete actions across branches, users and loa
      <div id="audit-table"></div>
    </div>

  </div>
</div>

<!-- Shared Modal -->
<div id="modal-overlay" class="modal-overlay" style="display:none" onclick="if(event.target===this)closeModal()">
  <div class="modal-box">
    <h3 id="modal-title"></h3>
    <div id="modal-body" class="modal-body"></div>
    <div class="msg" id="modal-msg" style="display:none"></div>
    <div class="modal-footer">
      <button class="modal-save" onclick="modalSave()">Save</button>
      <button class="modal-cancel" onclick="closeModal()">Cancel</button>
    </div>
  </div>
</div>
<script src="/static/app.js"></script>
<script>
let _branches = [], _borrowers = [], _staff = [], _admins = [];
let _modalCallback = null;
```

main | Launchpad | ⊗ 0 ⚠ 0 | You, yesterday | Ln 1, Col 1 | Spaces: 2 | UTF-8 | LF | HTML | Go Live | Prettier

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app > frontend > templates > admin > index.html > ...

```html
<html lang="en">
<body>

<!-- Shared Modal -->
<div id="modal-overlay" class="modal-overlay" style="display:none" onclick="if(event.target===this)closeModal()">
  <div class="modal-box">
    <h3 id="modal-title"></h3>
    <div id="modal-body" class="modal-body"></div>
    <div class="msg" id="modal-msg" style="display:none"></div>
    <div class="modal-footer">
      <button class="modal-save" onclick="modalSave()">Save</button>
      <button class="modal-cancel" onclick="closeModal()">Cancel</button>
    </div>
  </div>
</div>
<script src="/static/app.js"></script>
<script>
let _branches = [], _borrowers = [], _staff = [], _admins = [];
let _modalCallback = null;

function show(section) {
  document.querySelectorAll(".section").forEach(s => s.style.display = "none");
  document.querySelectorAll(".sidebar a").forEach(a => a.classList.remove("active"));
  document.getElementById("sec-" + section).style.display = "block";
  const nav = document.getElementById("nav-" + section);
  if (nav) nav.classList.add("active");
  if (section === "branches")  loadBranches();
  if (section === "borrowers") loadBorrowers();
  if (section === "staff")     loadStaff();
  if (section === "admins")    loadAdmins();
  if (section === "audit")     loadAuditLog();
}

/* — Modal core ———————————————————————— */
function openModal(title, bodyHTML, onSave) {
  document.getElementById("modal-title").textContent = title;
  document.getElementById("modal-body").innerHTML = bodyHTML;
  document.getElementById("modal-msg").style.display = "none";
  document.getElementById("modal-overlay").style.display = "flex";
  _modalCallback = onSave;
}
function closeModal() {
  document.getElementById("modal-overlay").style.display = "none";
  _modalCallback = null;
```

main | Launchpad | ⊗ 0 ⚠ 0 | You, yesterday | Ln 1, Col 1 | Spaces: 2 | UTF-8 | LF | HTML | Go Live | Prettier

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OUTLINE

TIMELINE

schema.sql | .gitignore | index.html

app > frontend > templates > admin > index.html > ...

```
2   <html lang="en">
9   <body>
185 <script>

142
143 /* — Branch select helper ———————————————— */
144 function branchOptions(selectedId = null) {
145   return _branches.map(b =>
146     `<option value="${b.branch_id}" ${selectedId == b.branch_id ? "selected" : ""}>${b.branch_name}</option>`
147   ).join("");
148 }
149
150 /* — Branches ———————————————————————— */
151 async function loadBranches() {
152   _branches = await (await apiFetch("/admin/branches")).json();
153   renderTable("branches-table", _branches, [
154     { key: "branch_name", label: "Branch Name" },
155     { key: "location",    label: "Location" },
156     { label: "Actions", render: r => `
157       <button class="btn btn-edit" onclick="openEditBranch(${r.branch_id})">Edit</button>
158       <button class="btn btn-red"  onclick="delBranch(${r.branch_id})">Delete</button>
159     `}
160   ]);
161 }
162
163 function branchForm(b = {}) {
164   return `
165     <label>Branch Name</label>
166     <input id="m-b-name" type="text" value="${b.branch_name || ""}" placeholder="e.g. Downtown Branch">
167     <label>Location</label>
168     <input id="m-b-loc" type="text" value="${b.location || ""}" placeholder="e.g. New York">
169   `;
170 }
171 function openAddBranch()     { openModal("New Branch",   branchForm(),  () => saveBranch(null)); }
172 function openEditBranch(id)  {
173   const b = _branches.find(x => x.branch_id == id);
174   openModal("Edit Branch", branchForm(b), () => saveBranch(id));
175 }
176 async function saveBranch(id) {
177   const name = document.getElementById("m-b-name").value.trim();
178   const loc  = document.getElementById("m-b-loc").value.trim();
179   if (!name || !loc) { modalMsg("Both fields are required.", true); return; }
180   const res = await apiFetch(id ? `/admin/branches/${id}` : "/admin/branches", {
181     method: id ? "PUT" : "POST", headers: authHeaders(),
```

main | Launchpad | ⊗ 0 ⚠ 0 | You, yesterday | Ln 1, Col 1 | Spaces: 2 | UTF-8 | LF | HTML | Go Live | Prettier

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schema.sql | .gitignore | index.html

app > frontend > templates > admin > index.html > ...

```
2   <html lang="en">
9   <body>
185 <script>
191 }
192
193 /* — Borrowers ———————————————————————— */
194 async function loadBorrowers() {
195   if (!_branches.length) _branches = await (await apiFetch("/admin/branches")).json();
196   _borrowers = await (await apiFetch("/admin/borrowers")).json();
197   renderTable("borrowers-table", _borrowers, [
198     { key: "name",        label: "Name" },
199     { key: "username",    label: "Username" },
200     { key: "phone",       label: "Phone" },
201     { key: "email",       label: "Email" },
202     { key: "branch_name", label: "Branch" },
203     { label: "Actions", render: r => `
204       <button class="btn btn-edit" onclick="openEditBorrower(${r.borrower_id})">Edit</button>
205       <button class="btn btn-red"  onclick="delBorrower(${r.borrower_id})">Delete</button>
206     `}
207   ]);
208 }
209
210 function borrowerForm(b = {}, isEdit = false) {
211   return `
212     ${!isEdit ? `
213     <div class="mrow">
214       <div><label>Username</label><input id="m-br-user" type="text" placeholder="username"></div>
215       <div><label>Password</label><input id="m-br-pass" type="password" placeholder="password"></div>
216     </div>` : ""}
217     <div class="mrow">
218       <div><label>Full Name</label><input id="m-br-name"  type="text"  value="${b.name  || ""}" placeholder="Full name"></d
219       <div><label>Email</label>    <input id="m-br-email" type="email" value="${b.email || ""}" placeholder="email@example.
220     </div>
221     <div class="mrow">
222       <div><label>Phone</label>        <input id="m-br-phone" type="text" value="${b.phone || ""}" placeholder="555-0101"><
223       <div><label>Date of Birth</label><input id="m-br-dob"   type="date" value="${b.dob   || ""}"></div>
224     </div>
225     <div class="mrow">
226       <div><label>PAN No</label><input id="m-br-pan" type="text" value="${b.pan_no || ""}" placeholder="ABCDE1234F"></d
227       <div><label>Branch</label><select id="m-br-branch">${branchOptions(b.branch_id)}</select></div>
228     </div>
229     <label>Address</label>
230     <input id="m-br-addr" type="text" value="${b.address || ""}" placeholder="123 Main St, City">
```

main | Launchpad | ⊗ 0 ⚠ 0 | You, yesterday | Ln 1, Col 1 | Spaces: 2 | UTF-8 | LF | HTML | Go Live | Prettier

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schema.sql | .gitignore | index.html

app > frontend > templates > admin > index.html > ...

```
2    <html lang="en">
9    <body>
185  <script>
238  async function saveBorrower(id) {
239    const isEdit = id !== null;
240    const body = {
241      name:     document.getElementById("m-br-name").value.trim(),
242      phone:    document.getElementById("m-br-phone").value.trim(),
243      address:  document.getElementById("m-br-addr").value.trim(),
244      email:    document.getElementById("m-br-email").value.trim(),
245      dob:      document.getElementById("m-br-dob").value,
246      pan_no:   document.getElementById("m-br-pan").value.trim(),
247      branch_id: parseInt(document.getElementById("m-br-branch").value)
248    };
249    if (!isEdit) {
250      body.username = document.getElementById("m-br-user").value.trim();
251      body.password = document.getElementById("m-br-pass").value;
252      if (!body.username || !body.password) { modalMsg("Username and password are required.", true); return; }
253    }
254    if (!body.name || !body.email) { modalMsg("Name and email are required.", true); return; }
255    const res = await apiFetch(isEdit ? `/admin/borrowers/${id}` : "/admin/borrowers", {
256      method: isEdit ? "PUT" : "POST", headers: authHeaders(), body: JSON.stringify(body)
257    });
258    if (res.ok) { closeModal(); loadBorrowers(); }
259    else { modalMsg((await res.json()).detail || "Failed.", true); }
260  }
261  async function delBorrower(id) {
262    if (!confirm("Delete this borrower and their login account?")) return;
263    await apiFetch("/admin/borrowers/" + id, { method: "DELETE", headers: authHeaders() });
264    loadBorrowers();
265  }
266
267  /* ── Staff ──────────────────────────────────────────── */
268  async function loadStaff() {
269    if (!_branches.length) _branches = await (await apiFetch("/admin/branches")).json();
270    _staff = await (await apiFetch("/admin/staff")).json();
271    renderTable("staff-table", _staff, [
272      { key: "name",        label: "Name" },
273      { key: "username",    label: "Username" },
274      { key: "branch_name", label: "Branch" },
275      { label: "Actions", render: r => `
276        <button class="btn btn-edit" onclick="openEditStaff('${r.staff_id}')">Edit</button>
277        <button class="btn btn-red" onclick="delStaff('${r.staff_id}')">Delete</button>
```

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app > frontend > templates > admin > index.html > ...

```
2    <html lang="en">
9    <body>
185  <script>
280  }
281
282  function staffForm(s = {}, isEdit = false) {
283    return `
284      ${!isEdit ? `
285      <div class="mrow">
286        <div><label>Username</label><input id="m-st-user" type="text" placeholder="username"></div>
287        <div><label>Password</label><input id="m-st-pass" type="password" placeholder="password"></div>
288      </div>` : ""}
289      <label>Full Name</label>
290      <input id="m-st-name" type="text" value="${s.name || ""}" placeholder="Full name">
291      <label>Branch</label>
292      <select id="m-st-branch">${branchOptions(s.branch_id)}</select>
293    `;
294  }
295  function openAddStaff()      { openModal("New Staff Member",  staffForm({}, false), () => saveStaff(null)); }
296  function openEditStaff(id)   {
297    const s = _staff.find(x => x.staff_id == id);
298    openModal("Edit Staff Member", staffForm(s, true), () => saveStaff(id));
299  }
300  async function saveStaff(id) {
301    const isEdit = id !== null;
302    const body = {
303      name:     document.getElementById("m-st-name").value.trim(),
304      branch_id: parseInt(document.getElementById("m-st-branch").value)
305    };
306    if (!isEdit) {
307      body.username = document.getElementById("m-st-user").value.trim();
308      body.password = document.getElementById("m-st-pass").value;
309      if (!body.username || !body.password) { modalMsg("Username and password are required.", true); return; }
310    }
311    if (!body.name) { modalMsg("Name is required.", true); return; }
312    const res = await apiFetch(isEdit ? `/admin/staff/${id}` : "/admin/staff", {
313      method: isEdit ? "PUT" : "POST", headers: authHeaders(), body: JSON.stringify(body)
314    });
315    if (res.ok) { closeModal(); loadStaff(); }
316    else { modalMsg((await res.json()).detail || "Failed.", true); }
317  }
318  async function delStaff(id) {
319    if (!confirm("Delete this staff member?")) return;
320    await apiFetch("/admin/staff/" + id, { method: "DELETE", headers: authHeaders() });
```

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OUTLINE

TIMELINE

schema.sql | .gitignore | index.html

app > frontend > templates > admin > index.html > ...

```
2   <html lang="en">
9   <body>
185 <script>
324 /* — Admins ———————————————————————————— */
325 async function loadAdmins() {
326   _admins = await (await apiFetch("/admin/admins")).json();
327   renderTable("admins-table", _admins, [
328     { key: "name",     label: "Name" },
329     { key: "username", label: "Username" },
330     { label: "Actions", render: r => `
331       <button class="btn btn-edit" onclick="openEditAdmin('${r.admin_id}')">Edit</button>
332       <button class="btn btn-red"  onclick="delAdmin('${r.admin_id}')">Delete</button>
333     `}
334   ]);
335 }
336
337 function adminForm(a = {}, isEdit = false) {
338   return `
339     ${!isEdit ? `
340     <div class="mrow">
341       <div><label>Username</label><input id="m-ad-user" type="text" placeholder="username"></div>
342       <div><label>Password</label><input id="m-ad-pass" type="password" placeholder="password"></div>
343     </div>` : ""}
344     <label>Full Name</label>
345     <input id="m-ad-name" type="text" value="${a.name || ""}" placeholder="Full name">
346   `;
347 }
348 function openAddAdmin()    { openModal("New Admin",   adminForm({}, false), () => saveAdmin(null)); }
349 function openEditAdmin(id) {
350   const a = _admins.find(x => x.admin_id == id);
351   openModal("Edit Admin", adminForm(a, true), () => saveAdmin(id));
352 }
353 async function saveAdmin(id) {
354   const isEdit = id !== null;
355   const body = { name: document.getElementById("m-ad-name").value.trim() };
356   if (!isEdit) {
357     body.username = document.getElementById("m-ad-user").value.trim();
358     body.password = document.getElementById("m-ad-pass").value;
359     if (!body.username || !body.password) { modalMsg("Username and password are required.", true); return; }
360   }
361   if (!body.name) { modalMsg("Name is required.", true); return; }
362   const res = await apiFetch(isEdit ? `/admin/admins/${id}` : "/admin/admins", {
363     method: isEdit ? "PUT" : "POST", headers: authHeaders(), body: JSON.stringify(body)
```

main | Launchpad | ⊗ 0 ⚠ 0 | You, yesterday | Ln 1, Col 1 | Spaces: 2 | UTF-8 | LF | HTML | Go Live | Prettier

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OUTLINE

TIMELINE

schema.sql | .gitignore | index.html

app > frontend > templates > admin > index.html > ...

```
2   <html lang="en">
9   <body>
185 <script>
353 async function saveAdmin(id) {
367 }
368 async function delAdmin(id) {
369   if (!confirm("Delete this admin account?")) return;
370   await apiFetch("/admin/admins/" + id, { method: "DELETE", headers: authHeaders() });
371   loadAdmins();
372 }
373
374 /* — Audit Log ———————————————————————————— */
375 function auditActionBadge(action) {
376   const map = { INSERT: "badge-active", UPDATE: "badge-pending", DELETE: "badge-rejected",
377                 APPROVE: "badge-completed", REJECT: "badge-rejected", REQUEST: "badge-pending" };
378   return `<span class="badge ${map[(action||"").toUpperCase()] || "badge-completed"}">${action||""}</span>`;
379 }
380 async function loadAuditLog() {
381   const data = await (await apiFetch("/admin/audit-logs")).json();
382   renderTable("audit-table", data, [
383     { key: "change_date", label: "Date" },
384     { key: "changed_by",  label: "By" },
385     { label: "Action",    render: r => auditActionBadge(r.action) },
386     { key: "note",        label: "Details" }
387   ]);
388 }
389
390 if (requireRole("admin")) {
391   show("branches");
392 }
393 </script>
394 </body>
395 </html>
396
```

main | Launchpad | ⊗ 0 ⚠ 0 | You, yesterday | Ln 1, Col 1 | Spaces: 2 | UTF-8 | LF | HTML | Go Live | Prettier

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OUTLINE

TIMELINE

schema.sql | .gitignore | index.html

app > frontend > templates > borrower > index.html > ...

You, yesterday | 1 author (You)

```html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Borrower</title>
<link rel="stylesheet" href="/static/style.css">
</head>
<body>
<div class="layout">
  <div class="sidebar">
    <h3>Borrower</h3>
    <a href="#" onclick="show('active')"><svg viewBox="0 0 24 24"><rect x="1" y="4" width="22" height="16" rx="2" ry="2"/><
    <a href="#" onclick="show('settled')"><svg viewBox="0 0 24 24"><path d="M22 11.08V12a10 10 0 11-5.93-9.14"/><polyline p
    <a href="#" onclick="show('request')"><svg viewBox="0 0 24 24"><circle cx="12" cy="12" r="10"/><line x1="12" y1="8" x2=
    <a href="#" onclick="show('payment')"><svg viewBox="0 0 24 24"><line x1="12" y1="1" x2="12" y2="23"/><path d="M17 5H9.5
    <a href="#" onclick="show('history')"><svg viewBox="0 0 24 24"><circle cx="12" cy="12" r="10"/><polyline points="12 6 1
    <button class="logout" onclick="logout()">Logout</button>
  </div>
  <div class="content">

    <!-- Active Loans -->
    <div id="sec-active" class="section" style="display:none">
      <h2>Active Loans</h2>
      <div id="active-loans-table"></div>
      <div id="active-loan-detail-panel" class="detail-panel" style="display:none"></div>
    </div>

    <!-- Settled Loans -->
    <div id="sec-settled" class="section" style="display:none">
      <h2>Settled Loans</h2>
      <div id="settled-loans-table"></div>
      <div id="settled-loan-detail-panel" class="detail-panel" style="display:none"></div>
    </div>

    <!-- Request Loan -->
    <div id="sec-request" class="section" style="display:none">
      <h2>Request a Loan</h2>
      <div class="request-grid">
        <div class="form-block request-card">
          <label>Amount</label><input id="rl-amount" type="number" min="1" oninput="markCalculationDirty()">
          <label>Loan Type</label>
          <select id="rl-type" onchange="markCalculationDirty()">
```

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OUTLINE

TIMELINE

schema.sql | .gitignore | index.html

app > frontend > templates > borrower > index.html > ...

```html
<html lang="en">
<body>
<div class="layout">
  <div class="content">
    <div id="sec-request" class="section" style="display:none">
            <th>Rule</th>
            <th>Effect</th>
          </tr>
        </thead>
        <tbody>
          <tr>
            <td>Personal loan</td>
            <td>Base 12.0%</td>
          </tr>
          <tr>
            <td>Home loan</td>
            <td>Base 8.5%</td>
          </tr>
          <tr>
            <td>Auto loan</td>
            <td>Base 9.5%</td>
          </tr>
          <tr>
            <td>Business loan</td>
            <td>Base 13.0%</td>
          </tr>
          <tr>
            <td>Tenure above 36 months</td>
            <td>+1.0%</td>
          </tr>
          <tr>
            <td>Amount above 500000</td>
            <td>+0.5%</td>
          </tr>
          <tr>
            <td>One successful prior tenure</td>
            <td>-0.1% once</td>
          </tr>
        </tbody>
      </table>
      <p class="rules-note">The final rate is calculated automatically by the system before a request is submitted.
    </div>
  </div>
```

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app > frontend > templates > borrower > index.html > ...

```html
<html lang="en">
<body>
<div class="layout">
  <div class="content">
    <div id="sec-payment" class="section" style="display:none">
      <div class="form-block payment-card">
        <p class="payment-intro">Pick a loan, choose when to pay, then confirm payment mode.</p>
        <label for="pay-loan">Select Active Loan</label>
        <select id="pay-loan" onchange="refreshPaymentView()"></select>

        <div class="payment-stats">
          <div class="stat-box">
            <div class="stat-label">EMIs Paid</div>
            <div class="stat-value" id="pay-stat-paid">0 / 0</div>
          </div>
          <div class="stat-box">
            <div class="stat-label">Amount Left</div>
            <div class="stat-value" id="pay-stat-left">0.00</div>
          </div>
        </div>

        <div class="payment-meta" id="pay-meta">
          <div><strong>Loan Start Date:</strong> —</div>
          <div><strong>Pay Before:</strong> —</div>
          <div><strong>Expected End Date:</strong> —</div>
          <div><strong>Approved By:</strong> —</div>
          <div><strong>Progress:</strong> <span id="pay-progress-text">0%</span></div>
          <div class="progress-track"><div id="pay-progress-bar" class="progress-bar"></div></div>
        </div>

        <div class="field-title">Payment Plan</div>
        <div class="plan-radios">
          <label class="radio-option"><input type="radio" name="pay-plan" value="current" checked onchange="refreshPayment
          <label class="radio-option"><input type="radio" name="pay-plan" value="next" onchange="refreshPaymentView()"> N
          <label class="radio-option"><input type="radio" name="pay-plan" value="custom" onchange="refreshPaymentView()">
        </div>

        <div id="pay-custom-wrap" class="custom-wrap" style="display:none">
          <label for="pay-emi-select">Select EMI Deadline</label>
          <select id="pay-emi-select" onchange="refreshPaymentView()"></select>
          <p class="rules-note">Selects all pending EMIs up to and including the chosen deadline.</p>
        </div>
```

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OUTLINE

TIMELINE

schema.sql | .gitignore | index.html

app > frontend > templates > borrower > index.html > ...

```html
<html lang="en">
<body>
<div class="layout">
  <div class="content">
    <div id="sec-payment" class="section" style="display:none">
        </div>
        <button onclick="submitPayment()">Pay Now</button>
      </div>
      <div class="msg" id="pay-msg"></div>
      <div id="pay-receipt" class="receipt-card" style="display:none"></div>
    </div>

    <div class="payment-rules">
      <h3>Quick Amount Guide</h3>
      <div class="plan-cards">
        <div class="plan-card">
          <div class="plan-name">Current Month</div>
          <div class="plan-amount" id="plan-one">—</div>
        </div>
        <div class="plan-card">
          <div class="plan-name">Next Month</div>
          <div class="plan-amount" id="plan-three">—</div>
        </div>
        <div class="plan-card">
          <div class="plan-name">Total Remaining</div>
          <div class="plan-amount" id="plan-full">—</div>
        </div>
      </div>
      <p class="rules-note">Custom lets you pick a due-date target and auto-includes all pending EMIs up to that target.
    </div>
  </div>
</div>

<!-- Payment History -->
<div id="sec-history" class="section" style="display:none">
  <h2>Payment History</h2>
  <div id="history-table"></div>
</div>

  </div>
</div>
<script src="/static/app.js"></script>
<script>
```

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OUTLINE

TIMELINE

schema.sql | .gitignore | index.html

app > frontend > templates > borrower > index.html > ...

```
2    <html lang="en">
9    <body>
194  <script>
200
201  function show(section) {
202    document.querySelectorAll(".section").forEach(s => s.style.display = "none");
203    document.getElementById("sec-" + section).style.display = "block";
204    if (section === "active") loadActiveLoans();
205    if (section === "settled") loadSettledLoans();
206    if (section === "payment") loadPaymentOptions();
207    if (section === "history") loadHistory();
208  }
209
210  async function loadActiveLoans() {
211    const res = await apiFetch("/borrower/loans");
212    const data = await res.json();
213    const active = data.filter(l => l.status !== "Completed" && l.status !== "Rejected");
214    renderTable("active-loans-table", active, [
215      { key: "loan_type",     label: "Type" },
216      { label: "Amount",      render: r => fmtCurrency(r.amount) },
217      { key: "interest_rate", label: "Rate (%)" },
218      { label: "Status",      render: r => statusBadge(r.status) },
219      { label: "EMIs Paid",   render: r => r.paid_emis + " / " + r.total_emis },
220      { label: "Details",     render: r => `<button class="btn" onclick="toggleLoanDetails(${r.loan_id}, 'active-loan-detail-
221    ]);
222  }
223
224  async function loadSettledLoans() {
225    const res = await apiFetch("/borrower/loans");
226    const data = await res.json();
227    const settled = data.filter(l => l.status === "Completed" || l.status === "Rejected");
228    renderTable("settled-loans-table", settled, [
229      { key: "loan_type",     label: "Type" },
230      { label: "Amount",      render: r => fmtCurrency(r.amount) },
231      { key: "interest_rate", label: "Rate (%)" },
232      { label: "Status",      render: r => statusBadge(r.status) },
233      { label: "EMIs Paid",   render: r => r.paid_emis + " / " + r.total_emis },
234      { label: "Details",     render: r => `<button class="btn" onclick="toggleLoanDetails(${r.loan_id}, 'settled-loan-de
235    ]);
236  }
237
238  function fmtDate(value) {
239    return value || "-";
```

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schema.sql | .gitignore | index.html

app > frontend > templates > borrower > index.html > ...

```
2    <html lang="en">
9    <body>
194  <script>
241
242  async function toggleLoanDetails(loanId, panelId) {
243    const panel = document.getElementById(panelId);
244    if (expandedLoanId === loanId) {
245      expandedLoanId = null;
246      panel.style.display = "none";
247      return;
248    }
249    expandedLoanId = loanId;
250    const res = await apiFetch(`/borrower/loans/${loanId}/details`);
251    const data = await res.json();
252    if (!res.ok) {
253      panel.style.display = "block";
254      panel.innerHTML = `<p>${data.detail || "Unable to load loan details."}</p>`;
255      return;
256    }
257    panel.style.display = "block";
258    panel.innerHTML = `
259      <h3>Loan ${data.loan_id} Details</h3>
260      <div class="detail-grid">
261        <div><strong>Branch:</strong> ${data.branch_name}</div>
262        <div><strong>Status:</strong> ${data.status}</div>
263        <div><strong>Approved By:</strong> ${data.approved_by || "Pending"}</div>
264        <div><strong>Interest Rate:</strong> ${fmtRate(data.interest_rate)}%</div>
265        <div><strong>Monthly Installment:</strong> ${fmtAmount(data.monthly_installment)}</div>
266        <div><strong>Total Payment:</strong> ${fmtAmount(data.total_payment)}</div>
267        <div><strong>Loan Start Date:</strong> ${fmtDate(data.start_date)}</div>
268        <div><strong>Pay Before:</strong> ${fmtDate(data.next_due_date)}</div>
269        <div><strong>Expected End Date:</strong> ${fmtDate(data.expected_end_date)}</div>
270        <div><strong>EMIs Paid:</strong> ${data.paid_emis} / ${data.total_emis}</div>
271        <div><strong>Amount Left:</strong> ${fmtAmount(data.remaining_amount)}</div>
272        <div><strong>Progress:</strong> ${data.progress_percent}%</div>
273      </div>
274    `;
275  }
276
277  function getLoanRequestBody() {
278    return {
279      amount: Number.parseFloat(document.getElementById("rl-amount").value),
280      loan_type: document.getElementById("rl-type").value,
281      tenure_months: Number.parseInt(document.getElementById("rl-tenure").value)
```

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schema.sql | .gitignore | index.html

app > frontend > templates > borrower > index.html > ...

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242 async function toggleLoanDetails(loanId, panelId) {
274   ;
275 }
276
277 function getLoanRequestBody() {
278   return {
279     amount: Number.parseFloat(document.getElementById("rl-amount").value),
280     loan_type: document.getElementById("rl-type").value,
281     tenure_months: Number.parseInt(document.getElementById("rl-tenure").value)
282   };
283 }
284
285 function getLoanRequestKey(body) {
286   return JSON.stringify(body);
287 }
288
289 function markCalculationDirty() {
290   latestRate = null;
291   lastCalculatedKey = null;
292   document.getElementById("rl-submit").disabled = true;
293   document.getElementById("rl-msg").textContent = "";
294   document.getElementById("rl-rate-preview").textContent = "Enter loan details to calculate your interest rate.";
295   document.getElementById("rl-repayment-preview").innerHTML = `
296     <div><strong>Monthly Installment:</strong> —</div>
297     <div><strong>Total Payment:</strong> —</div>
298   `;
299 }
300
301 async function calculateInterest() {
302   const body = getLoanRequestBody();
303   const res = await apiFetch("/borrower/loans/calculate", { method: "POST", headers: authHeaders(), body: JSON.stringify(bo
304   const data = await res.json();
305   if (!res.ok) {
306     document.getElementById("rl-rate-preview").textContent = "Unable to calculate interest rate.";
307     document.getElementById("rl-msg").textContent = "Error.";
308     document.getElementById("rl-submit").disabled = true;
309     latestRate = null;
310     return;
311   }
312   latestRate = data.interest_rate;
313   lastCalculatedKey = getLoanRequestKey(body);
```

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OUTLINE

TIMELINE

schema.sql | .gitignore | index.html

app > frontend > templates > borrower > index.html > ...

```
2   <html lang="en">
9   <body>
194 <script>
301 async function calculateInterest() {
312   latestRate = data.interest_rate;
313   lastCalculatedKey = getLoanRequestKey(body);
314   document.getElementById("rl-rate-preview").textContent = data.successful_tenure_discount
315     ? `Applied interest rate: ${data.interest_rate}% (includes one-time 0.1% successful tenure discount)`
316     : `Applied interest rate: ${data.interest_rate}%`;
317   document.getElementById("rl-repayment-preview").innerHTML = `
318     <div><strong>Monthly Installment:</strong> ${fmtAmount(data.monthly_installment)}</div>
319     <div><strong>Total Payment:</strong> ${fmtAmount(data.total_payment)}</div>
320   `;
321   document.getElementById("rl-msg").textContent = "Rate calculated. Submit when ready.";
322   document.getElementById("rl-submit").disabled = false;
323 }
324
325 async function submitLoan() {
326   const body = getLoanRequestBody();
327   if (latestRate === null || lastCalculatedKey !== getLoanRequestKey(body)) {
328     document.getElementById("rl-msg").textContent = "Calculate interest first.";
329     return;
330   }
331   const res = await apiFetch("/borrower/loans/request", { method: "POST", headers: authHeaders(), body: JSON.stringify(body
332   const data = await res.json();
333   document.getElementById("rl-msg").textContent = res.ok
334     ? `Loan request submitted at ${data.interest_rate}% interest. Monthly installment ${fmtAmount(data.monthly_installment)
335     : "Error.";
336   if (res.ok) {
337     document.getElementById("rl-amount").value = "";
338     document.getElementById("rl-type").value = "Personal";
339     document.getElementById("rl-tenure").value = "";
340     document.getElementById("rl-rate-preview").textContent = "Enter loan details to calculate your interest rate.";
341     document.getElementById("rl-repayment-preview").innerHTML = `<div><strong>Monthly Installment:</strong> —</div><div><st
342     document.getElementById("rl-submit").disabled = true;
343     latestRate = null;
344     lastCalculatedKey = null;
345   }
346 }
347
348 function getSelectedPaymentOption() {
349   const loanId = Number.parseInt(document.getElementById("pay-loan").value);
350   return paymentOptions.find(o => o.loan_id === loanId);
351 }
```

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TIMELINE

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app > frontend > templates > borrower > index.html > ...

```
<html lang="en">
<body>
<script>

function getSelectedPaymentOption() {
  const loanId = Number.parseInt(document.getElementById("pay-loan").value);
  return paymentOptions.find(o => o.loan_id === loanId);
}

function getSelectedPlan() {
  return (document.querySelector('input[name="pay-plan"]:checked') || {}).value || "current";
}

function _parseDue(dateStr) {
  const [y, m, d] = dateStr.split("-").map(Number);
  return { year: y, month: m, day: d };
}

const _MON = ["Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct","Nov","Dec"];

function refreshPaymentView() {
  const loan = getSelectedPaymentOption();
  if (!loan) return;

  document.getElementById("pay-stat-paid").textContent = `${loan.paid_emis} / ${loan.total_emis}`;
  document.getElementById("pay-stat-left").textContent = fmtAmount(loan.remaining_amount);
  document.getElementById("pay-meta").innerHTML =
    <div><strong>Loan Start Date:</strong> ${fmtDate(loan.start_date)}</div>
    <div><strong>Pay Before:</strong> ${fmtDate(loan.next_due_date)}</div>
    <div><strong>Expected End Date:</strong> ${fmtDate(loan.expected_end_date)}</div>
    <div><strong>Approved By:</strong> ${loan.approved_by || "Pending"}</div>
    <div><strong>Progress:</strong> <span id="pay-progress-text">${loan.progress_percent}%</span></div>
    <div class="progress-track"><div id="pay-progress-bar" class="progress-bar" style="width:${loan.progress_percent}%"></d
  `;

  const now = new Date();
  const cm = now.getMonth() + 1, cy = now.getFullYear();
  const nm = cm === 12 ? 1 : cm + 1, ny = cm === 12 ? cy + 1 : cy;

  const pendingEmis = loan.pending_emis || [];
  const currEmi = pendingEmis.find(e => { const p = _parseDue(e.due_date); return p.month === cm && p.year === cy; });
  const nextEmi = pendingEmis.find(e => { const p = _parseDue(e.due_date); return p.month === nm && p.year === ny; });
```

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TIMELINE

schema.sql | .gitignore | index.html

app > frontend > templates > borrower > index.html > ...

```
<html lang="en">
<body>
<script>
function refreshPaymentView() {

  document.getElementById("plan-one").textContent   = currEmi ? fmtAmount(currEmi.amount_due) : `${fmtAmount(0)} (paid)`;
  document.getElementById("plan-three").textContent = nextEmi ? fmtAmount(nextEmi.amount_due) : `${fmtAmount(0)} (paid)`;
  document.getElementById("plan-full").textContent  = fmtAmount(loan.full_amount);

  const plan = getSelectedPlan();
  const customWrap = document.getElementById("pay-custom-wrap");
  const emiSelect  = document.getElementById("pay-emi-select");

  if (plan === "custom") {
    customWrap.style.display = "block";
    const previousSelection = Number.parseInt(emiSelect.value, 10) || 0;
    const emiSignature = pendingEmis.map(e => e.emi_number).join(",");

    // Repopulate when loan OR pending EMI set changes.
    if (emiSelect.dataset.loanId !== String(loan.loan_id) || emiSelect.dataset.emiSignature !== emiSignature) {
      emiSelect.innerHTML = pendingEmis.map(e => {
        const p = _parseDue(e.due_date);
        return `<option value="${e.emi_number}">EMI #${e.emi_number}  |  ${p.day} ${_MON[p.month - 1]} ${p.year}</option>`;
      }).join("");
      emiSelect.dataset.loanId = loan.loan_id;
      emiSelect.dataset.emiSignature = emiSignature;

      // Preserve valid previous selection, else fallback to last EMI for "pay till end" convenience.
      const hasPrevious = pendingEmis.some(e => e.emi_number === previousSelection);
      if (hasPrevious) {
        emiSelect.value = String(previousSelection);
      } else if (pendingEmis.length) {
        emiSelect.value = String(pendingEmis[pendingEmis.length - 1].emi_number);
      }
    }

    const selNum = Number.parseInt(emiSelect.value, 10) || 0;
    const idx = pendingEmis.findIndex(e => e.emi_number === selNum);
    const relevant = idx >= 0 ? pendingEmis.slice(0, idx + 1) : [];
    _paySelectedAmount = Math.round(relevant.reduce((s, e) => s + e.amount_due, 0) * 100) / 100;
  } else {
    customWrap.style.display = "none";
    const target = plan === "current" ? currEmi : nextEmi;
```

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OUTLINE

TIMELINE

schema.sql | .gitignore | index.html

app > frontend > templates > borrower > index.html > ...

```
2    <html lang="en">
9    <body>
194  <script>
364  function refreshPaymentView() {
427
428    document.getElementById("pay-amount").value = fmtAmount(_paySelectedAmount);
429  }
430
431  function fmtAmount(value) {
432    return fmtCurrency(value);
433  }
434
435  function fmtRate(value) {
436    return Number(value || 0).toLocaleString("en-BD", { minimumFractionDigits: 2, maximumFractionDigits: 2 });
437  }
438
439  async function loadPaymentOptions() {
440    const res = await apiFetch("/borrower/payment-options");
441    const data = await res.json();
442    paymentOptions = data;
443    const sel = document.getElementById("pay-loan");
444    const msg = document.getElementById("pay-msg");
445    if (!paymentOptions.length) {
445      sel.innerHTML = "<option>No active loan with pending EMIs</option>";
447      msg.textContent = "No pending EMI to pay.";
448      document.getElementById("pay-amount").value = "";
449      document.getElementById("plan-one").textContent = "-";
450      document.getElementById("plan-three").textContent = "-";
451      document.getElementById("plan-full").textContent = "-";
452      document.getElementById("pay-stat-paid").textContent = "0 / 0";
453      document.getElementById("pay-stat-left").textContent = fmtAmount(0);
454      document.getElementById("pay-meta").innerHTML = `
455        <div><strong>Loan Start Date:</strong> —</div>
456        <div><strong>Pay Before:</strong> —</div>
457        <div><strong>Expected End Date:</strong> —</div>
458        <div><strong>Approved By:</strong> —</div>
459        <div><strong>Progress:</strong> <span id="pay-progress-text">0%</span></div>
460        <div class="progress-track"><div id="pay-progress-bar" class="progress-bar"></div></div>
461      `;
462      document.getElementById("pay-receipt").style.display = "none";
463      return;
464    }
465    sel.innerHTML = paymentOptions.map(o => `<option value="${o.loan_id}">Loan ${o.loan_id}</option>`).join("");
```

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TIMELINE

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app > frontend > templates > borrower > index.html > ...

```
2    <html lang="en">
9    <body>
194  <script>
439  async function loadPaymentOptions() {
466    msg.textContent = "";
467    refreshPaymentView();
468  }
469
470  async function submitPayment() {
471    const loan = getSelectedPaymentOption();
472    if (!loan) return;
473    if (_paySelectedAmount <= 0) {
474      document.getElementById("pay-msg").textContent = "No pending EMI for this period.";
475      return;
476    }
477    const body = {
478      loan_id: loan.loan_id,
479      plan: "custom",
480      payment_mode: document.getElementById("pay-mode").value,
481      custom_amount: _paySelectedAmount
482    };
483    const res = await apiFetch("/borrower/payments", { method: "POST", headers: authHeaders(), body: JSON.stringify(body) });
484    const data = await res.json();
485    document.getElementById("pay-msg").textContent = res.ok
486      ? `Payment recorded. Paid ${data.count_paid} EMI(s), amount ${fmtAmount(data.amount_paid)}.`
487      : (data.detail || "Error.");
488    if (res.ok) {
489      const receipt = data.receipt;
490      const receiptEl = document.getElementById("pay-receipt");
491      receiptEl.style.display = "block";
492      receiptEl.innerHTML = `
493        <h4>Payment Receipt</h4>
494        <div class="detail-grid">
495          <div><strong>Receipt ID:</strong> ${receipt.payment_id}</div>
496          <div><strong>Date:</strong> ${fmtDate(receipt.payment_date)}</div>
497          <div><strong>Loan ID:</strong> ${receipt.loan_id}</div>
498          <div><strong>Plan:</strong> ${receipt.plan}</div>
499          <div><strong>Payment Mode:</strong> ${receipt.payment_mode}</div>
500          <div><strong>Amount Paid:</strong> ${fmtAmount(data.amount_paid)}</div>
501          <div><strong>Paid Until EMI:</strong> ${receipt.paid_until_emi}</div>
502          <div><strong>Loan Start Date:</strong> ${fmtDate(receipt.loan_start_date)}</div>
503          <div><strong>Pay Before:</strong> ${fmtDate(receipt.pay_before_date)}</div>
504          <div><strong>Expected End Date:</strong> ${fmtDate(receipt.expected_end_date)}</div>
```

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OUTLINE

TIMELINE

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app > frontend > templates > borrower > index.html > ...

```html
<html lang="en">
<body>
<script>
async function submitPayment() {
        <div><strong>Amount Paid:</strong> ${fmtAmount(data.amount_paid)}</div>
        <div><strong>Paid Until EMI:</strong> ${receipt.paid_until_emi}</div>
        <div><strong>Loan Start Date:</strong> ${fmtDate(receipt.loan_start_date)}</div>
        <div><strong>Pay Before:</strong> ${fmtDate(receipt.pay_before_date)}</div>
        <div><strong>Expected End Date:</strong> ${fmtDate(receipt.expected_end_date)}</div>
        <div><strong>EMIs Paid:</strong> ${receipt.paid_emis} / ${receipt.total_emis}</div>
        <div><strong>Amount Left:</strong> ${fmtAmount(receipt.remaining_amount)}</div>
      </div>
      <div class="progress-track"><div class="progress-bar" style="width:${receipt.progress_percent}%"></div></div>
    `;
    loadPaymentOptions();
    loadActiveLoans();
  }
}

async function loadHistory() {
  const res = await apiFetch("/borrower/payments");
  const data = await res.json();
  renderTable("history-table", data, [
    { key: "payment_date",  label: "Date" },
    { label: "Loan #",       render: r => "Loan " + r.loan_id },
    { label: "Amount",       render: r => fmtCurrency(r.amount_paid) },
    { key: "payment_mode",  label: "Mode" }
  ]);
}

if (requireRole("borrower")) {
  show("active");
}
</script>
</body>
</html>
```

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You, yesterday | 1 author (You)

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Staff - Loan Manager</title>
  <link rel="stylesheet" href="/static/style.css">
</head>
<body>
<div class="layout">
  <div class="sidebar">
    <h3>Staff Panel</h3>
    <a href="#" id="nav-requests" onclick="show('requests')">
      <svg viewBox="0 0 24 24"><path d="M14 2H6a2 2 0 00-2 2v16a2 2 0 002 2h12a2 2 0 002-2V8z"/><polyline points="14 2 14 8
    </a>
    <a href="#" id="nav-users" onclick="show('users')">
      <svg viewBox="0 0 24 24"><path d="M20 21v-2a4 4 0 00-4-4H8a4 4 0 00-4 4v2"/><circle cx="12" cy="7" r="4"/></svg>Branc
    </a>
    <a href="#" id="nav-active" onclick="show('active')">
      <svg viewBox="0 0 24 24"><polyline points="22 12 18 12 15 21 9 3 6 12 2 12"/></svg>Active Loans
    </a>
    <a href="#" id="nav-archive" onclick="show('archive')">
      <svg viewBox="0 0 24 24"><polyline points="21 8 21 21 3 21 3 8"/><rect x="1" y="3" width="22" height="5"/><line x1="1
    </a>
    <a href="#" id="nav-audit" onclick="show('audit')">
      <svg viewBox="0 0 24 24"><path d="M9 5H7a2 2 0 00-2 2v12a2 2 0 002 2h10a2 2 0 002-2V7a2 2 0 00-2-2h-2"/><rect x="9" y=
    </a>
    <button class="logout" onclick="logout()">
      <svg viewBox="0 0 24 24" style="width:14px;height:14px;stroke:currentColor;fill:none;stroke-width:2;vertical-align:mi
    </button>
  </div>
  <div class="content">

    <!-- Loan Requests -->
    <div id="sec-requests" class="section" style="display:none">
      <h2>Pending Loan Requests</h2>
      <p class="section-meta">New loan applications waiting for your approval at this branch.</p>
      <div id="requests-table"></div>
    </div>

    <!-- Branch Borrowers -->
    <div id="sec-users" class="section" style="display:none">
      <h2>Branch Borrowers</h2>
```

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app > frontend > templates > staff > index.html > ...

```
2   <html lang="en">
9   <body>
10  <div class="layout">
32    <div class="content">
47
48      <!-- Active Loans -->
49      <div id="sec-active" class="section" style="display:none">
50        <h2>Active Loans</h2>
51        <p class="section-meta">Currently running loans — EMI progress and risk status at a glance.</p>
52        <div id="active-loans-table"></div>
53      </div>
54
55      <!-- Loan Archive -->
56      <div id="sec-archive" class="section" style="display:none">
57        <h2>Loan Archive</h2>
58        <p class="section-meta">Completed and rejected loans at this branch.</p>
59        <div id="archive-table"></div>
60      </div>
61
62      <!-- Audit Log -->
63      <div id="sec-audit" class="section" style="display:none">
64        <h2>Audit Log</h2>
65        <p class="section-meta">All entity changes related to this branch.</p>
66        <div id="audit-table"></div>
67      </div>
68
69    </div>
70  </div>
71  <script src="/static/app.js"></script>
72  <script>
73  function show(section) {
74    document.querySelectorAll(".section").forEach(s => s.style.display = "none");
75    document.querySelectorAll(".sidebar a").forEach(a => a.classList.remove("active"));
76    document.getElementById("sec-" + section).style.display = "block";
77    const nav = document.getElementById("nav-" + section);
78    if (nav) nav.classList.add("active");
79    if (section === "requests") loadRequests();
80    if (section === "users")    loadUsers();
81    if (section === "active")   loadActiveLoans();
82    if (section === "archive")  loadLoanArchive();
83    if (section === "audit")    loadAuditLog();
84  }
85
85  async function loadRequests() {
```

main | Launchpad | 0 0 | You, yesterday | Ln 1, Col 1 | Spaces: 2 | UTF-8 | LF | HTML | Go Live | Prettier

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OUTLINE

TIMELINE

schema.sql | .gitignore | index.html

app > frontend > templates > staff > index.html > ...

```
2   <html lang="en">
9   <body>
72  <script>
155
156
157 async function loadAuditLog() {
158   const data = await (await apiFetch("/staff/audit-logs")).json();
159   renderTable("audit-table", data, [
160     { key: "change_date", label: "Date" },
161     { key: "changed_by",  label: "By" },
162     { label: "Action",    render: r => auditActionBadge(r.action) },
163     { key: "note",        label: "Details" }
164   ]);
165 }
166
167 if (requireRole("staff")) {
168   show("requests");
169 }
170 </script>
171 </body>
172 </html>
173
```

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OUTLINE

TIMELINE

schema.sql | .gitignore | login.html

app > frontend > templates > login.html > ...

You, yesterday | 1 author (You)

```html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Login</title>
<link rel="stylesheet" href="/static/style.css">
</head>
<body>
<div class="login-wrap">
  <div class="login-box">
    <div class="login-brand">
      <div class="brand-icon">
        <svg viewBox="0 0 24 24"><path d="M3 9l9-7 9 7v11a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2z"/><polyline points="9 22 9 12 15 1
      </div>
      <h2>Loan Manager</h2>
      <p>Credit Risk &amp; Monitoring Platform</p>
    </div>
    <label for="role">Role</label>
    <select id="role">
      <option value="admin">Admin</option>
      <option value="staff">Staff</option>
      <option value="borrower">Borrower</option>
    </select>
    <label for="username">Username</label>
    <input type="text" id="username" placeholder="Username">
    <label for="password">Password</label>
    <input type="password" id="password" placeholder="Password">
    <button onclick="doLogin()">Login</button>
    <div class="err" id="err"></div>
  </div>
</div>
<script>
async function doLogin() {
  const role = document.getElementById("role").value;
  const username = document.getElementById("username").value;
  const password = document.getElementById("password").value;
  const res = await fetch("/auth/login", {
    method: "POST",
    headers: { "Content-Type": "application/json" },
    body: JSON.stringify({ username, password, role })
  });
  const data = await res.json();
```

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OUTLINE

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schema.sql | .gitignore | login.html

app > frontend > templates > login.html > ...

```html
<html lang="en">
<body>
<div class="login-wrap">
  <div class="login-box">
    <label for="role">Role</label>
    <select id="role">
      <option value="admin">Admin</option>
      <option value="staff">Staff</option>
      <option value="borrower">Borrower</option>
    </select>
    <label for="username">Username</label>
    <input type="text" id="username" placeholder="Username">
    <label for="password">Password</label>
    <input type="password" id="password" placeholder="Password">
    <button onclick="doLogin()">Login</button>
    <div class="err" id="err"></div>
  </div>
</div>
<script>
async function doLogin() {
  const role = document.getElementById("role").value;
  const username = document.getElementById("username").value;
  const password = document.getElementById("password").value;
  const res = await fetch("/auth/login", {
    method: "POST",
    headers: { "Content-Type": "application/json" },
    body: JSON.stringify({ username, password, role })
  });
  const data = await res.json();
  if (!res.ok) {
    document.getElementById("err").textContent = data.detail || "Login failed";
    return;
  }
  localStorage.setItem("token", data.token);
  localStorage.setItem("role", data.role);
  globalThis.location.href = "/" + data.role + "/";
}
</script>
</body>
</html>
```

main | Launchpad | 0 0 | You, yesterday | Ln 1, Col 1 | Spaces: 2 | UTF-8 | LF | HTML | Go Live | Prettier

Backend Development

The backend uses FastAPI and Python with a MySQL database
Database Connectivity: I implemented a centralized `query()` function in `db.py` that uses **parameterized queries** (`%s`) to prevent SQL injection attacks
Security & Auth: Developed `auth.py` to handle password hashing via **bcrypt** and session management using **UUID** tokens stored in an in-memory dictionary.
Business Logic: Coded the `staff.py` logic for loan approval and risk recalculation, as well as the simple interest calculation for EMIs in `borrower.py`.

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app > python > routers > admin.py > admin_home

You, yesterday | 1 author (You)

```python
from fastapi import APIRouter, Depends, Request, HTTPException
from fastapi.templating import Jinja2Templates
from pathlib import Path
from pydantic import BaseModel
from db import query
from auth import get_current_user, hash_password, new_id
from audit import write_audit_log

router = APIRouter()
templates = Jinja2Templates(directory=str(Path(__file__).parent.parent.parent / "frontend" / "templates"))


def delete_entity_with_user(table, id_col, id_val, user):
    """Helper: delete entity and its user account, with audit logging"""
    old = query(f"SELECT * FROM {table} WHERE {id_col} = %s", (id_val,))
    if old:
        query(f"DELETE FROM {table} WHERE {id_col} = %s", (id_val,), fetch=False)
        query("DELETE FROM User WHERE user_id = %s", (old[0]["user_id"],), fetch=False)
        write_audit_log(table, id_val, "DELETE", user=user, note="{} deleted".format(table))
    return {"ok": True}


You, yesterday | 1 author (You)
class BranchBody(BaseModel):
    branch_name: str
    location: str

You, yesterday | 1 author (You)
class BorrowerBody(BaseModel):
    username: str
    password: str
    name: str
    phone: str
    address: str
    email: str
    dob: str
    pan_no: str
    branch_id: int

You, yesterday | 1 author (You)
class StaffBody(BaseModel):
    username: str
    password: str
```

main | Launchpad | ⊗ 5 ⚠ 0 | You, yesterday | tasnubatabassumahona (1 day ago) | Ln 55, Col 34 | Spaces: 4 | UTF-8 | LF | Python | Python 3.9.6 | Go Live | Prettier

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OUTLINE

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You, yesterday | 1 author (You)

```python
class AdminBody(BaseModel):
    username: str
    password: str
    name: str


from auth import require_role
admin_only = require_role("admin")


@router.get("/")
def admin_home(request: Request):        You, yesterday • Final Version
    return templates.TemplateResponse(request, "admin/index.html")


@router.get("/branches")
def list_branches(user=Depends(admin_only)):
    return query("SELECT * FROM Branch")

@router.post("/branches")
def create_branch(body: BranchBody, user=Depends(admin_only)):
    branch_id = query("INSERT INTO Branch (branch_name, location) VALUES (%s, %s)", (body.branch_name, body.location), fetc
    write_audit_log("Branch", branch_id, "INSERT", user=user, note="Branch '{}' added".format(body.branch_name))
    return {"ok": True}

@router.delete("/branches/{branch_id}")
def delete_branch(branch_id: int, user=Depends(admin_only)):
    old = query("SELECT * FROM Branch WHERE branch_id = %s", (branch_id,))
    if old:
        query("DELETE FROM Branch WHERE branch_id = %s", (branch_id,), fetch=False)
        write_audit_log("Branch", branch_id, "DELETE", user=user, note="Branch '{}' deleted".format(old[0].get("branch_name'
    return {"ok": True}


@router.get("/borrowers")
def list_borrowers(user=Depends(admin_only)):
    return query("""
        SELECT b.*, u.username, br.branch_name
        FROM   Borrower AS b
        LEFT JOIN User   AS u  ON b.user_id   = u.user_id
        LEFT JOIN Branch AS br ON b.branch_id = br.branch_id
```

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app > python > routers > admin.py > admin_home

```python
@router.post("/borrowers")
def create_borrower(body: BorrowerBody, user=Depends(admin_only)):
    uid = new_id()
    hashed = hash_password(body.password)
    query("INSERT INTO User (user_id, username, password_hash, role) VALUES (%s, %s, %s, 'borrower')",
          (uid, body.username, hashed), fetch=False)
    borrower_id = query(
        "INSERT INTO Borrower (user_id, name, phone, address, email, dob, pan_no, branch_id) VALUES (%s,%s,%s,%s,%s,%s,%s,%s
        (uid, body.name, body.phone, body.address, body.email, body.dob, body.pan_no, body.branch_id),
        fetch=False)
    write_audit_log("Borrower", borrower_id, "INSERT", user=user, note="Borrower '{}' added".format(body.username))
    return {"ok": True}

@router.delete("/borrowers/{borrower_id}")
def delete_borrower(borrower_id: int, user=Depends(admin_only)):
    return delete_entity_with_user("Borrower", "borrower_id", borrower_id, user)


@router.get("/staff")
def list_staff(user=Depends(admin_only)):
    return query("""
        SELECT s.*, u.username, b.branch_name
        FROM   Staff AS s
        INNER JOIN User   AS u ON s.user_id   = u.user_id
        LEFT  JOIN Branch AS b ON s.branch_id = b.branch_id
    """)

@router.post("/staff")
def create_staff(body: StaffBody, user=Depends(admin_only)):
    uid, sid = new_id(), new_id()
    hashed = hash_password(body.password)
    query("INSERT INTO User (user_id, username, password_hash, role) VALUES (%s, %s, %s, 'staff')",
          (uid, body.username, hashed), fetch=False)
    query("INSERT INTO Staff (staff_id, user_id, name, branch_id) VALUES (%s, %s, %s, %s)",
          (sid, uid, body.name, body.branch_id), fetch=False)
    write_audit_log("Staff", sid, "INSERT", user=user, note="Staff '{}' added".format(body.username))
    return {"ok": True}

@router.delete("/staff/{staff_id}")
def delete_staff(staff_id: str, user=Depends(admin_only)):
    return delete_entity_with_user("Staff", "staff_id", staff_id, user)
```

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app > python > routers > admin.py > admin_home

```python
@router.get("/admins")
def list_admins(user=Depends(admin_only)):
    return query("""
        SELECT a.*, u.username
        FROM   Admin AS a
        INNER JOIN User AS u ON a.user_id = u.user_id
    """)

@router.post("/admins")
def create_admin(body: AdminBody, user=Depends(admin_only)):
    uid, aid = new_id(), new_id()
    hashed = hash_password(body.password)
    query("INSERT INTO User (user_id, username, password_hash, role) VALUES (%s, %s, %s, 'admin')",
          (uid, body.username, hashed), fetch=False)
    query("INSERT INTO Admin (admin_id, user_id, name) VALUES (%s, %s, %s)",
          (aid, uid, body.name), fetch=False)
    write_audit_log("Admin", aid, "INSERT", user=user, note="Admin '{}' added".format(body.username))
    return {"ok": True}

@router.delete("/admins/{admin_id}")
def delete_admin(admin_id: str, user=Depends(admin_only)):
    return delete_entity_with_user("Admin", "admin_id", admin_id, user)


# — Update (PUT) endpoints ————————————————————————————

You, yesterday | 1 author (You)
class BranchUpdateBody(BaseModel):
    branch_name: str
    location: str

You, yesterday | 1 author (You)
class BorrowerUpdateBody(BaseModel):
    name: str
    phone: str
    address: str
    email: str
    dob: str
    pan_no: str
    branch_id: int

You, yesterday | 1 author (You)
class StaffUpdateBody(BaseModel):
    name: str
```

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```python

@router.put("/branches/{branch_id}")
def update_branch(branch_id: int, body: BranchUpdateBody, user=Depends(admin_only)):
    old = query("SELECT * FROM Branch WHERE branch_id = %s", (branch_id,))
    if not old:
        raise HTTPException(status_code=404, detail="Branch not found")
    query("UPDATE Branch SET branch_name = %s, location = %s WHERE branch_id = %s",
          (body.branch_name, body.location, branch_id), fetch=False)
    write_audit_log("Branch", branch_id, "UPDATE", user=user, note="Branch '{}' updated".format(body.branch_name))
    return {"ok": True}

@router.put("/borrowers/{borrower_id}")
def update_borrower(borrower_id: int, body: BorrowerUpdateBody, user=Depends(admin_only)):
    old = query("SELECT * FROM Borrower WHERE borrower_id = %s", (borrower_id,))
    if not old:
        raise HTTPException(status_code=404, detail="Borrower not found")
    query("""
        UPDATE Borrower
        SET    name = %s, phone = %s, address = %s, email = %s,
               dob = %s, pan_no = %s, branch_id = %s
        WHERE  borrower_id = %s
    """, (body.name, body.phone, body.address, body.email,
          body.dob, body.pan_no, body.branch_id, borrower_id), fetch=False)
    write_audit_log("Borrower", borrower_id, "UPDATE", user=user, note="Borrower '{}' updated".format(body.name))
    return {"ok": True}

@router.put("/staff/{staff_id}")
def update_staff(staff_id: str, body: StaffUpdateBody, user=Depends(admin_only)):
    old = query("SELECT * FROM Staff WHERE staff_id = %s", (staff_id,))
    if not old:
        raise HTTPException(status_code=404, detail="Staff not found")
    query("UPDATE Staff SET name = %s, branch_id = %s WHERE staff_id = %s",
          (body.name, body.branch_id, staff_id), fetch=False)
    write_audit_log("Staff", staff_id, "UPDATE", user=user, note="Staff '{}' updated".format(body.name))
    return {"ok": True}

@router.put("/admins/{admin_id}")
def update_admin(admin_id: str, body: AdminUpdateBody, user=Depends(admin_only)):
    old = query("SELECT * FROM Admin WHERE admin_id = %s", (admin_id,))
    if not old:
        raise HTTPException(status_code=404, detail="Admin not found")
    query("UPDATE Admin SET name = %s WHERE admin_id = %s",
          (body.name, admin_id), fetch=False)
```

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```python
    return {"ok": True}

@router.put("/admins/{admin_id}")
def update_admin(admin_id: str, body: AdminUpdateBody, user=Depends(admin_only)):
    old = query("SELECT * FROM Admin WHERE admin_id = %s", (admin_id,))
    if not old:
        raise HTTPException(status_code=404, detail="Admin not found")
    query("UPDATE Admin SET name = %s WHERE admin_id = %s",
          (body.name, admin_id), fetch=False)
    write_audit_log("Admin", admin_id, "UPDATE", user=user, note="Admin '{}' updated".format(body.name))
    return {"ok": True}


@router.get("/audit-logs")
def audit_logs(user=Depends(admin_only)):
    return query("""
        SELECT change_date, changed_by, table_name, action, new_value AS note
        FROM   Audit_Log
        ORDER BY log_id DESC
        LIMIT 200
    """)

```

main | Launchpad | 5 0 | You, yesterday | tasnubatabassumahona (1 day ago) | Ln 55, Col 34 | Spaces: 4 | UTF-8 | LF | Python | Python 3.9.6 | Go Live | Prettier

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schema.sql | borrower.py 9+ | .gitignore

app > python > routers > borrower.py > ...

```python
router = APIRouter()
templates = Jinja2Templates(directory=str(Path(__file__).parent.parent.parent / "frontend" / "templates"))
_approved_by_exists = None


You, yesterday | 1 author (You)
class LoanRequestBody(BaseModel):
    amount: float
    loan_type: str
    tenure_months: int
        You, yesterday • Final Version ...
You, yesterday | 1 author (You)
class PaymentBody(BaseModel):
    loan_id: int
    plan: str
    payment_mode: str
    custom_amount: Optional[float] = None


from auth import require_role
borrower_only = require_role("borrower")

def get_borrower_id(user_id):
    rows = query("SELECT borrower_id FROM Borrower WHERE user_id = %s", (user_id,))
    return rows[0]["borrower_id"] if rows else None


def has_successful_tenure(borrower_id):
    rows = query("SELECT 1 FROM Loan WHERE borrower_id = %s AND status = 'Completed' LIMIT 1", (borrower_id,))
    return bool(rows)


def calculate_interest_rate(loan_type, amount, tenure_months, successful_tenure):
    base_rates = {
        "Personal": 12.0,
        "Home": 8.5,
        "Auto": 9.5,
        "Business": 13.0,
    }

    rate = base_rates.get(loan_type, 12.0)

    if tenure_months > 36:
```

main | Launchpad | ⊗ 68 ⚠ 0 | You, yesterday | tasnubatabassumahona (1 day ago) | Ln 21, Col 1 | Spaces: 4 | UTF-8 | LF | Python | Python 3.9.6 | Go Live | Prettier

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def calculate_interest_rate(loan_type, amount, tenure_months, successful_tenure):
    base_rates = {
        "Personal": 12.0,
        "Home": 8.5,
        "Auto": 9.5,
        "Business": 13.0,
    }

    rate = base_rates.get(loan_type, 12.0)

    if tenure_months > 36:
        rate += 1.0

    if amount > 500000:
        rate += 0.5

    if successful_tenure:
        rate -= 0.1

    return round(rate, 2)


def calculate_repayment_summary(amount, interest_rate, tenure_months):
    total_payment = round(amount * (1 + interest_rate / 100), 2)
    monthly_installment = round(total_payment / tenure_months, 2)
    return monthly_installment, total_payment


def has_approved_by_column():
    global _approved_by_exists
    if _approved_by_exists is not None:
        return _approved_by_exists
    rows = query("""
        SELECT COUNT(*) AS c FROM INFORMATION_SCHEMA.COLUMNS
        WHERE TABLE_SCHEMA = DATABASE() AND TABLE_NAME = 'Loan' AND COLUMN_NAME = 'approved_by'
    """)
    _approved_by_exists = bool(rows[0]["c"])
    return _approved_by_exists


def get_loan_progress_summary(loan_id):
    emis = query("""
```

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```python
def calculate_repayment_summary(amount, interest_rate, tenure_months):
    total_payment = round(amount * (1 + interest_rate / 100), 2)
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    return monthly_installment, total_payment


def has_approved_by_column():
    global _approved_by_exists
    if _approved_by_exists is not None:
        return _approved_by_exists
    rows = query("""
        SELECT COUNT(*) AS c FROM INFORMATION_SCHEMA.COLUMNS
        WHERE TABLE_SCHEMA = DATABASE() AND TABLE_NAME = 'Loan' AND COLUMN_NAME = 'approved_by'
    """)
    _approved_by_exists = bool(rows[0]["c"])
    return _approved_by_exists


def get_loan_progress_summary(loan_id):
    emis = query("""
        SELECT due_date, amount_due, status
        FROM EMI_Schedule WHERE loan_id = %s ORDER BY emi_number
    """, (loan_id,))
    paid = sum(1 for e in emis if e["status"] == "Paid")
    total = len(emis)
    pending = [e for e in emis if e["status"] == "Pending"]
    remaining_amount = round(sum(e["amount_due"] for e in pending), 2)
    progress_percent = round((paid / total) * 100, 1) if total else 0.0
    next_due = pending[0]["due_date"] if pending else None
    end_due = emis[-1]["due_date"] if emis else None
    return {
        "paid_emis": paid, "total_emis": total, "remaining_emis": max(total - paid, 0),
        "remaining_amount": remaining_amount, "next_due_date": str(next_due) if next_due else None,
        "expected_end_date": str(end_due) if end_due else None, "progress_percent": progress_percent
    }


def get_approved_by_name(loan_id):
    if not has_approved_by_column():
        return None
    rows = query("""
        SELECT u.username AS approved_by_name
```

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app > python > routers > borrower.py > ...

```python
def get_approved_by_name(loan_id):
    if not has_approved_by_column():
        return None
    rows = query("""
        SELECT u.username AS approved_by_name
        FROM   Loan AS l
        LEFT JOIN Staff AS s ON l.approved_by = s.staff_id
        LEFT JOIN User  AS u ON s.user_id     = u.user_id
        WHERE  l.loan_id = %s
    """, (loan_id,))
    return rows[0]["approved_by_name"] if rows and rows[0]["approved_by_name"] else None


def get_pending_emis(loan_id):
    return query("""
        SELECT emi_id, emi_number, due_date, amount_due
        FROM   EMI_Schedule
        WHERE  loan_id = %s
          AND  status = 'Pending'
        ORDER BY emi_number
    """, (loan_id,))


def get_payment_option_details(borrower_id):
    loans = query("""
        SELECT loan_id, start_date
        FROM   Loan
        WHERE  borrower_id = %s
          AND  status = 'Active'
    """, (borrower_id,))
    result = []
    for loan in loans:
        loan_id = loan["loan_id"]
        pending = get_pending_emis(loan_id)
        if not pending:
            continue
        summary = get_loan_progress_summary(loan_id)
        one_month_amount = round(pending[0]["amount_due"], 2)
        three_month_amount = round(sum(e["amount_due"] for e in pending[:3]), 2)
        result.append({
            "loan_id": loan_id,
            "start date": str(loan["start date"]) if loan["start date"] else None,
```

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```python
def get_payment_option_details(borrower_id):
    loans = query("""
        SELECT loan_id, start_date
        FROM   Loan
        WHERE  borrower_id = %s
          AND  status = 'Active'
    """, (borrower_id,))
    result = []
    for loan in loans:
        loan_id = loan["loan_id"]
        pending = get_pending_emis(loan_id)
        if not pending:
            continue
        summary = get_loan_progress_summary(loan_id)
        one_month_amount = round(pending[0]["amount_due"], 2)
        three_month_amount = round(sum(e["amount_due"] for e in pending[:3]), 2)
        result.append({
            "loan_id": loan_id,
            "start_date": str(loan["start_date"]) if loan["start_date"] else None,
            "approved_by": get_approved_by_name(loan_id),
            "paid_emis": summary["paid_emis"],
            "total_emis": summary["total_emis"],
            "remaining_emis": summary["remaining_emis"],
            "remaining_amount": summary["remaining_amount"],
            "next_due_date": summary["next_due_date"],
            "expected_end_date": summary["expected_end_date"],
            "progress_percent": summary["progress_percent"],
            "one_month_amount": one_month_amount,
            "three_month_amount": three_month_amount,
            "full_amount": summary["remaining_amount"],
            "pending_emis": [
                {"emi_number": e["emi_number"], "due_date": str(e["due_date"]), "amount_due": float(e["amount_due"])}
                for e in pending
            ]
        })
    return result


@router.get("/")
def borrower_home(request: Request):
    return templates.TemplateResponse(request, "borrower/index.html")
```

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app > python > routers > borrower.py > ...

```python
@router.get("/loans")
def my_loans(user=Depends(borrower_only)):
    bid = get_borrower_id(user["user_id"])
    loans = query("""
        SELECT l.*, br.branch_name
        FROM   Loan AS l
        INNER JOIN Branch AS br ON l.branch_id = br.branch_id
        WHERE  l.borrower_id = %s
    """, (bid,))
    for loan in loans:
        summary = get_loan_progress_summary(loan["loan_id"])
        loan["paid_emis"] = summary["paid_emis"]
        loan["total_emis"] = summary["total_emis"]
        loan["remaining_amount"] = summary["remaining_amount"]
        loan["progress_percent"] = summary["progress_percent"]
        loan["approved_by"] = get_approved_by_name(loan["loan_id"])
    return loans


@router.get("/loans/{loan_id}/details")
def loan_details(loan_id: int, user=Depends(borrower_only)):
    bid = get_borrower_id(user["user_id"])
    rows = query("""
        SELECT l.loan_id, l.amount, l.interest_rate, l.start_date, l.loan_type, l.tenure_months, l.status,
               br.branch_name
        FROM   Loan AS l
        INNER JOIN Branch AS br ON l.branch_id = br.branch_id
        WHERE  l.loan_id = %s
          AND  l.borrower_id = %s
    """, (loan_id, bid))
    if not rows:
        raise HTTPException(status_code=404, detail="Loan not found")
    loan = rows[0]
    summary = get_loan_progress_summary(loan_id)
    monthly_installment, total_payment = calculate_repayment_summary(
        loan["amount"], loan["interest_rate"], loan["tenure_months"]
    )
    loan["approved_by"] = get_approved_by_name(loan_id)
    loan["monthly_installment"] = monthly_installment
    loan["total_payment"] = total_payment
    loan.update(summary)
    return loan
```

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app > python > routers > borrower.py > ...

```python
    return loan

@router.post("/loans/calculate")
def calculate_loan_interest(body: LoanRequestBody, user=Depends(borrower_only)):
    bid = get_borrower_id(user["user_id"])
    if not bid:
        raise HTTPException(status_code=404, detail="Borrower not found")
    successful_tenure = has_successful_tenure(bid)
    interest_rate = calculate_interest_rate(
        body.loan_type,
        body.amount,
        body.tenure_months,
        successful_tenure
    )
    monthly_installment, total_payment = calculate_repayment_summary(body.amount, interest_rate, body.tenure_months)
    return {
        "interest_rate": interest_rate,
        "successful_tenure_discount": successful_tenure,
        "monthly_installment": monthly_installment,
        "total_payment": total_payment
    }

@router.post("/loans/request")
def request_loan(body: LoanRequestBody, user=Depends(borrower_only)):
    rows = query("SELECT borrower_id, branch_id FROM Borrower WHERE user_id = %s", (user["user_id"],))
    if not rows:
        raise HTTPException(status_code=404, detail="Borrower not found")
    bid = rows[0]["borrower_id"]
    branch_id = rows[0]["branch_id"]
    interest_rate = calculate_interest_rate(
        body.loan_type,
        body.amount,
        body.tenure_months,
        has_successful_tenure(bid)
    )
    monthly_installment, total_payment = calculate_repayment_summary(body.amount, interest_rate, body.tenure_months)
    loan_id = query("""
        INSERT INTO Loan (borrower_id, branch_id, amount, interest_rate, loan_type, tenure_months, status)
        VALUES (%s, %s, %s, %s, %s, %s, 'Pending')
    """, (bid, branch_id, body.amount, interest_rate, body.loan_type, body.tenure_months), fetch=False)
    write_audit_log("Loan", loan_id, "REQUEST", user=user, note="Loan requested: {} for {} months".format(body.loan_typ
    return {
        "ok": True,
```

main | Launchpad | 68 | 0 | You, yesterday | tasnubatabassumahona (1 day ago) | Ln 21, Col 1 | Spaces: 4 | UTF-8 | LF | Python | Python 3.9.6 | Go Live | Prettier

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app > python > routers > borrower.py > ...

```python

@router.get("/payment-options")
def payment_options(user=Depends(borrower_only)):
    bid = get_borrower_id(user["user_id"])
    return get_payment_option_details(bid)

@router.post("/payments")
def make_payment(body: PaymentBody, user=Depends(borrower_only)):
    bid = get_borrower_id(user["user_id"])
    if not query("SELECT loan_id FROM Loan WHERE loan_id = %s AND borrower_id = %s", (body.loan_id, bid)):
        raise HTTPException(status_code=403, detail="Not your loan")

    pending = get_pending_emis(body.loan_id)
    if not pending:
        raise HTTPException(status_code=400, detail="No pending EMI for this loan")

    plan = body.plan.lower().strip()
    selected_emis = []

    if plan == "one":
        selected_emis = pending[:1]
    elif plan == "three":
        selected_emis = pending[:3]
    elif plan == "full":
        selected_emis = pending
    elif plan == "custom":
        if body.custom_amount is None or body.custom_amount <= 0:
            raise HTTPException(status_code=400, detail="custom_amount must be greater than 0 for custom plan")
        running = 0.0
        target = round(body.custom_amount, 2)
        for emi in pending:
            next_running = round(running + emi["amount_due"], 2)
            if next_running <= target + 0.01:
                selected_emis.append(emi)
                running = next_running
                if abs(running - target) <= 0.01:
                    break
            else:
                break
        if not selected_emis or abs(round(sum(e["amount_due"] for e in selected_emis), 2) - target) > 0.01:
            raise HTTPException(status_code=400, detail="Custom amount must match an exact sum of earliest pending EMIs",
    else:
        raise HTTPException(status_code=400, detail="Invalid payment plan")
```

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app > python > routers > borrower.py > ...

```python
263 def make_payment(body: PaymentBody, user=Depends(borrower_only)):
298         raise HTTPException(status_code=400, detail="Invalid payment plan")
299
300     amount_paid = round(sum(e["amount_due"] for e in selected_emis), 2)
301     paid_until_emi = selected_emis[-1]["emi_number"]
302     payment_id = query("""
303         INSERT INTO Payment (loan_id, payment_date, amount_paid, payment_mode)
304         VALUES (%s, %s, %s, %s)
305     """, (body.loan_id, date.today(), amount_paid, body.payment_mode), fetch=False)
306     write_audit_log("Payment", payment_id, "INSERT", user=user, note="Payment {} via {} on Loan #{}".format(amount_paid, bo
307
308     for emi in selected_emis:
309         query("INSERT INTO Payment_Allocation (payment_id, emi_id, allocated_amount) VALUES (%s, %s, %s)",
310               (payment_id, emi["emi_id"], emi["amount_due"]), fetch=False)
311         query("UPDATE EMI_Schedule SET status = 'Paid' WHERE emi_id = %s", (emi["emi_id"],), fetch=False)
312
313     remaining = query("SELECT 1 FROM EMI_Schedule WHERE loan_id = %s AND status != 'Paid' LIMIT 1", (body.loan_id,))
314     if not remaining:
315         query("UPDATE Loan SET status = 'Completed' WHERE loan_id = %s", (body.loan_id,), fetch=False)
316         write_audit_log("Loan", body.loan_id, "UPDATE", user=user, note="Loan #{} completed".format(body.loan_id))
317     upsert_risk(body.loan_id)
318
319     details = query("SELECT start_date FROM Loan WHERE loan_id = %s", (body.loan_id,))[0]
320     summary = get_loan_progress_summary(body.loan_id)
321     return {
322         "ok": True,
323         "amount_paid": amount_paid,
324         "count_paid": len(selected_emis),
325         "receipt": {
326             "payment_id": payment_id,
327             "payment_date": str(date.today()),
328             "loan_id": body.loan_id,
329             "plan": plan,
330             "payment_mode": body.payment_mode,
331             "paid_until_emi": paid_until_emi,
332             "loan_start_date": str(details["start_date"]) if details["start_date"] else None,
333             "pay_before_date": summary["next_due_date"],
334             "expected_end_date": summary["expected_end_date"],
335             "paid_emis": summary["paid_emis"],
336             "total_emis": summary["total_emis"],
337             "remaining_amount": summary["remaining_amount"],
338             "progress_percent": summary["progress_percent"]
339         }
340     }
```

main | Launchpad | 68 0 | You, yesterday | tasnubatabassumahona (1 day ago) | Ln 21, Col 1 | Spaces: 4 | UTF-8 | LF | Python | Python 3.9.6 | Go Live | Prettier

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OUTLINE

TIMELINE

schema.sql | borrower.py 9+ | .gitignore

app > python > routers > borrower.py > ...

```python
263 def make_payment(body: PaymentBody, user=Depends(borrower_only)):
336             "total_emis": summary["total_emis"],
337             "remaining_amount": summary["remaining_amount"],
338             "progress_percent": summary["progress_percent"]
339         }
340     }
341
342 @router.get("/payments")
343 def payment_history(user=Depends(borrower_only)):
344     bid = get_borrower_id(user["user_id"])
345     return query("""
346         SELECT p.payment_id, p.loan_id, p.payment_date, p.amount_paid, p.payment_mode
347         FROM   Payment AS p
348         INNER JOIN Loan AS l ON p.loan_id = l.loan_id
349         WHERE  l.borrower_id = %s
350         ORDER BY p.payment_date DESC
351     """, (bid,))
352
```

main | Launchpad | 68 0 | You, yesterday | tasnubatabassumahona (1 day ago) | Ln 21, Col 1 | Spaces: 4 | UTF-8 | LF | Python | Python 3.9.6 | Go Live | Prettier

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OUTLINE

TIMELINE

schema.sql | staff.py 9+ | .gitignore

app > python > routers > staff.py > ...

You, yesterday | 1 author (You)

```python
from fastapi import APIRouter, Depends, Request, HTTPException
from fastapi.templating import JinjaTemplates
from pathlib import Path
from pydantic import BaseModel
from datetime import date, timedelta
from db import query
from auth import get_current_user
from audit import write_audit_log

router = APIRouter()
templates = JinjaTemplates(directory=str(Path(__file__).parent.parent.parent / "frontend" / "templates"))
_approved_by_ready = False


You, yesterday | 1 author (You)
class LoanActionBody(BaseModel):
    loan_id: int


from auth import require_role
staff_only = require_role("staff")


def classify_risk(loan_id):
    emis = query("""
        SELECT due_date
        FROM   EMI_Schedule
        WHERE  loan_id = %s
          AND  status != 'Paid'
        ORDER BY due_date
    """, (loan_id,))
    if not emis:
        return 1
    today = date.today()
    max_days = max(
        ((today - e["due_date"]).days for e in emis if e["due_date"] < today),
        default=0
    )
    if max_days == 0:  return 1
    if max_days <= 30: return 2
    if max_days <= 60: return 3
    return 4
```

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app > python > routers > staff.py > ...

```python


from auth import require_role
staff_only = require_role("staff")


def classify_risk(loan_id):
    emis = query("""
        SELECT due_date
        FROM   EMI_Schedule
        WHERE  loan_id = %s
          AND  status != 'Paid'
        ORDER BY due_date
    """, (loan_id,))
    if not emis:
        return 1
    today = date.today()
    max_days = max(
        ((today - e["due_date"]).days for e in emis if e["due_date"] < today),
        default=0
    )
    if max_days == 0:  return 1
    if max_days <= 30: return 2
    if max_days <= 60: return 3
    return 4

def upsert_risk(loan_id):
    bucket_id = classify_risk(loan_id)
    exists = query("SELECT risk_id FROM Loan_Risk_Status WHERE loan_id = %s", (loan_id,))
    if exists:
        query("""
            UPDATE Loan_Risk_Status
            SET    bucket_id = %s, assessed_date = %s
            WHERE  loan_id = %s
        """, (bucket_id, date.today(), loan_id), fetch=False)
    else:
        query("""
            INSERT INTO Loan_Risk_Status (loan_id, bucket_id, assessed_date)
            VALUES (%s, %s, %s)
        """, (loan_id, bucket_id, date.today()), fetch=False)


def ensure_approved_by_column():
    global _approved_by_ready
```

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app > python > routers > staff.py > ...

```python
43  def upsert_risk(loan_id):
55              VALUES (%s, %s, %s)
56          """, (loan_id, bucket_id, date.today()), fetch=False)
57
58
59  def ensure_approved_by_column():
60      global _approved_by_ready
61      if _approved_by_ready:
62          return
63      exists = query("""
64          SELECT COUNT(*) AS c
65          FROM   INFORMATION_SCHEMA.COLUMNS
66          WHERE  TABLE_SCHEMA = DATABASE()
67            AND  TABLE_NAME = 'Loan'
68            AND  COLUMN_NAME = 'approved_by'
69      """)[0]["c"]
70      if not exists:
71          query("ALTER TABLE Loan ADD COLUMN approved_by CHAR(16) NULL", fetch=False)
72      col_type = query("""
73          SELECT DATA_TYPE AS t
74          FROM   INFORMATION_SCHEMA.COLUMNS
75          WHERE  TABLE_SCHEMA = DATABASE()
76            AND  TABLE_NAME = 'Loan'
77            AND  COLUMN_NAME = 'approved_by'
78          LIMIT 1
79      """)[0]["t"]
80      if col_type.lower() != "char":
81          query("ALTER TABLE Loan MODIFY COLUMN approved_by CHAR(16) NULL", fetch=False)
82      fk_exists = query("""
83          SELECT COUNT(*) AS c
84          FROM   INFORMATION_SCHEMA.KEY_COLUMN_USAGE
85          WHERE  TABLE_SCHEMA = DATABASE()
86            AND  TABLE_NAME = 'Loan'
87            AND  COLUMN_NAME = 'approved_by'
88            AND  REFERENCED_TABLE_NAME = 'Staff'
89      """)[0]["c"]
90      if not fk_exists:
91          query("ALTER TABLE Loan ADD CONSTRAINT fk_loan_approved_by FOREIGN KEY (approved_by) REFERENCES Staff(staff_id)
92      _approved_by_ready = True
93
94
95  @router.get("/")
96  def staff_home(request: Request):
```

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TIMELINE

schema.sql | staff.py 9+ | .gitignore

app > python > routers > staff.py > ...

```python
92      _approved_by_ready = True
93
94
95  @router.get("/")
96  def staff_home(request: Request):
97      return templates.TemplateResponse(request, "staff/index.html")
98
99
100 @router.get("/loan-requests")
101 def loan_requests(user=Depends(staff_only)):
102     return query("""
103         SELECT l.*, b.name AS borrower_name
104         FROM   Loan AS l
105         INNER JOIN Borrower AS b ON l.borrower_id = b.borrower_id
106         WHERE  l.branch_id = %s
107           AND  l.status = 'Pending'
108     """, (user["branch_id"],))
109
110 @router.post("/loan-requests/approve")
111 def approve_loan(body: LoanActionBody, user=Depends(staff_only)):
112     ensure_approved_by_column()
113     loans = query("SELECT * FROM Loan WHERE loan_id = %s AND branch_id = %s", (body.loan_id, user["branch_id"]))
114     if not loans:
115         raise HTTPException(status_code=404, detail="Loan not found")
116     loan = loans[0]
117     staff_rows = query("SELECT staff_id FROM Staff WHERE user_id = %s", (user["user_id"],))
118     if not staff_rows:
119         raise HTTPException(status_code=404, detail="Staff record not found")
120     staff_id = staff_rows[0]["staff_id"]
121     query("""
122         UPDATE Loan
123         SET    status = 'Active', start_date = %s, approved_by = %s
124         WHERE  loan_id = %s
125     """, (date.today(), staff_id, body.loan_id), fetch=False)
126     write_audit_log("Loan", body.loan_id, "APPROVE", user=user, note="Loan #{} approved".format(body.loan_id))
127     emi_amount = round(loan["amount"] * (1 + loan["interest_rate"] / 100) / loan["tenure_months"], 2)
128     for i in range(1, loan["tenure_months"] + 1):
129         query("""
130             INSERT INTO EMI_Schedule (loan_id, emi_number, due_date, amount_due, status)
131             VALUES (%s, %s, %s, %s, 'Pending')
132         """, (body.loan_id, i, date.today() + timedelta(days=30 * i), emi_amount), fetch=False)
133     upsert_risk(body.loan_id)
134     return {"ok": True}
```

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OUTLINE

TIMELINE

schema.sql | staff.py 9+ | .gitignore

app > python > routers > staff.py > ...

```python
    return {"ok": True}

@router.post("/loan-requests/reject")
def reject_loan(body: LoanActionBody, user=Depends(staff_only)):
    query("UPDATE Loan SET status = 'Rejected' WHERE loan_id = %s AND branch_id = %s",
          (body.loan_id, user["branch_id"]), fetch=False)
    write_audit_log("Loan", body.loan_id, "REJECT", user=user, note="Loan #{} rejected".format(body.loan_id))
    return {"ok": True}


@router.get("/branch-users")
def branch_users(user=Depends(staff_only)):
    return query("""
        SELECT 'borrower' AS role, b.borrower_id AS profile_id, u.username, b.name,
               b.email, b.phone
        FROM   Borrower AS b
        LEFT JOIN User AS u ON b.user_id = u.user_id
        WHERE  b.branch_id = %s
    """, (user["branch_id"],))


@router.get("/loan-monitoring")
def loan_monitoring(user=Depends(staff_only)):
    loans = query("""
        SELECT l.loan_id, l.borrower_id, l.amount, l.status, l.start_date,
               b.name AS borrower_name, rb.bucket_name
        FROM   Loan AS l
        INNER JOIN Borrower AS b ON l.borrower_id = b.borrower_id
        LEFT JOIN Loan_Risk_Status AS lrs ON l.loan_id = lrs.loan_id
        LEFT JOIN Risk_Bucket AS rb ON lrs.bucket_id = rb.bucket_id
        WHERE  l.branch_id = %s
          AND  l.status IN ('Active', 'Completed')
        ORDER BY l.loan_id DESC
    """, (user["branch_id"],))

    for loan in loans:
        paid = query("SELECT COUNT(*) AS c FROM EMI_Schedule WHERE loan_id = %s AND status = 'Paid'", (loan["loan_id"],
        total = query("SELECT COUNT(*) AS c FROM EMI_Schedule WHERE loan_id = %s", (loan["loan_id"],))[0]["c"]
        next_due = query("""
            SELECT due_date
            FROM   EMI_Schedule
            WHERE  loan_id = %s
              AND  status = 'Pending'
            ORDER BY emi_number
```

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schema.sql | staff.py 9+ | .gitignore

app > python > routers > staff.py > ...

```python
def loan_monitoring(user=Depends(staff_only)):
        loan["next_due_date"] = str(next_due[0]["due_date"]) if next_due else None

    return loans


@router.get("/active-loans")
def active_loans(user=Depends(staff_only)):
    loans = query("""
        SELECT l.loan_id, l.borrower_id, l.amount, l.status, l.start_date, l.loan_type,
               b.name AS borrower_name, rb.bucket_name
        FROM   Loan AS l
        INNER JOIN Borrower AS b ON l.borrower_id = b.borrower_id
        LEFT  JOIN Loan_Risk_Status AS lrs ON l.loan_id = lrs.loan_id
        LEFT  JOIN Risk_Bucket AS rb ON lrs.bucket_id = rb.bucket_id
        WHERE  l.branch_id = %s AND l.status = 'Active'
        ORDER BY l.loan_id DESC
    """, (user["branch_id"],))
    for loan in loans:
        paid  = query("SELECT COUNT(*) AS c FROM EMI_Schedule WHERE loan_id = %s AND status = 'Paid'",
                      (loan["loan_id"],))[0]["c"]
        total = query("SELECT COUNT(*) AS c FROM EMI_Schedule WHERE loan_id = %s",
                      (loan["loan_id"],))[0]["c"]
        next_due = query("""
            SELECT due_date FROM EMI_Schedule
            WHERE  loan_id = %s AND status = 'Pending'
            ORDER  BY emi_number LIMIT 1
        """, (loan["loan_id"],))
        loan["paid_emis"]     = paid
        loan["total_emis"]    = total
        loan["next_due_date"] = str(next_due[0]["due_date"]) if next_due else None
    return loans


@router.get("/loan-archive")
def loan_archive(user=Depends(staff_only)):
    return query("""
        SELECT l.loan_id, l.amount, l.status, l.start_date, l.loan_type,
               b.name AS borrower_name
        FROM   Loan AS l
        INNER JOIN Borrower AS b ON l.borrower_id = b.borrower_id
        WHERE  l.branch_id = %s AND l.status IN ('Completed', 'Rejected')
        ORDER BY l.loan_id DESC
```

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app > python > routers > staff.py > ...

```python
def loan_archive(user=Depends(staff_only)):
        SELECT l.loan_id, l.amount, l.status, l.start_date, l.loan_type,
               b.name AS borrower_name
        FROM   Loan AS l
        INNER JOIN Borrower AS b ON l.borrower_id = b.borrower_id
        WHERE  l.branch_id = %s AND l.status IN ('Completed', 'Rejected')
        ORDER BY l.loan_id DESC
    """, (user["branch_id"],))


@router.get("/audit-logs")
def staff_audit_logs(user=Depends(staff_only)):
    username_rows = query("SELECT username FROM User WHERE user_id = %s", (user["user_id"],))
    username = username_rows[0]["username"] if username_rows else ""
    return query("""
        SELECT change_date, changed_by, table_name, action, new_value AS note
        FROM   Audit_Log
        WHERE  changed_by = %s
           OR  (table_name = 'Loan' AND record_id IN (
                   SELECT loan_id FROM Loan WHERE branch_id = %s))
           OR  (table_name = 'Payment' AND record_id IN (
                   SELECT p.payment_id FROM Payment AS p
                   JOIN   Loan AS l ON p.loan_id = l.loan_id
                   WHERE  l.branch_id = %s))
        ORDER BY log_id DESC
        LIMIT 200
    """, (username, user["branch_id"], user["branch_id"]))
```

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app > python > audit.py > ...

```python
from datetime import date
from db import query


def _username(user):
    if not user:
        return "system"
    rows = query("SELECT username FROM User WHERE user_id = %s", (user.get("user_id"),))
    return rows[0]["username"] if rows else "unknown"


def write_audit_log(table_name, record_id, action, user=None, note=""):
    query(
        "INSERT INTO Audit_Log (table_name, record_id, action, changed_by, change_date, new_value) "
        "VALUES (%s, %s, %s, %s, %s, %s)",
        (table_name, record_id, action, _username(user), date.today(), note),
        fetch=False,
    )
```

Source Code Repository

https://github.com/ttahona/DBMS_project-Credit-risk-and-Loan-monitoring-Platform-

Conclusion

The Credit Risk & Loan Monitoring Platform successfully demonstrates how relational database design supports complex financial operations. Through proper normalization and a secure backend architecture, the system provides a reliable environment for managing loans, tracking repayments, and auditing user actions in a multi-branch banking scenario.

References

https://youtu.be/d5Hf6d6QtIo?si=8yl9OgiutcNj180P
https://youtu.be/J8QPm8_j6lM?si=tXGlc2v_0wLIIg9_